

Original source families: covariance angles, rational detection and complete observed heat

Split-Zero research programme

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The source weights vary in the complete original covariance. Every normal-source cross term, original relation, physical coordinate, observation map and minimum section remains in the calculation. The first new bound controls the change of the quotient angle without an upper bound for the covariance norm or rank:

$$|\delta(\mathbf{y}) - \delta(\mathbf{x})| \leq \min(\dim K, \dim V) [\max(0, \mathbf{y} - \mathbf{x}) - \min(0, \mathbf{y} - \mathbf{x})].$$

Polynomial changes of positive weights therefore cost $O(k \log q)$ in the original four-cutoff angle. The actual initial angle is retained.

The second result gives the exact coefficient inverse for all proper rational sources on $Jd - v \leq q'$, including repeated or colliding poles. Its original-metric lower bound passes to an entire slow singular space with the sharp observed fraction in UFR14. UFR20 proves the coordinate map needed for applying the physical S -conductor to a y -filter.

The full observed positive heat and the squared observed action have different operators. UH1-32 proves their exact memory relation and the uniform trace estimate with every leakage and memory block:

$$\sup_{t \geq 0} \left| \text{Tr}(\Lambda e^{-tM^\dagger M} L) - \text{Tr} e^{-tM_B^\dagger M_B} \right| \leq \max(\text{rank } D, \text{rank } B_H) \leq 8k - 16.$$

For the actual joint source, UFR21-23 evaluates the norm comparison and proves an $o(kq)$ determinant-pencil error throughout the displayed regularizer and source-weight ranges.

The actual activated adjacent source has two columns: its outgoing scalar column and the residual of that same column against the full joint source. TC1-29 proves their complete determinant identity, retains the complex cross term, and bounds its logarithm by $O(q)$ uniformly in activation. This proves the $o(kq)$ angle change at its actual activated baseline.

The complete received 87-equation source is reproduced at the end. The correction after its (72) is explicit: the small positive semidefinite matrix has the *squared* singular values, including zeros. Its strict finite guard proves positivity when used. Complete earlier proof sources and original human equation sources accompany the reader. None of these results assigns the remaining canonical kernel coefficient or native complex projected-current sign.

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The exact projection and memory mechanisms

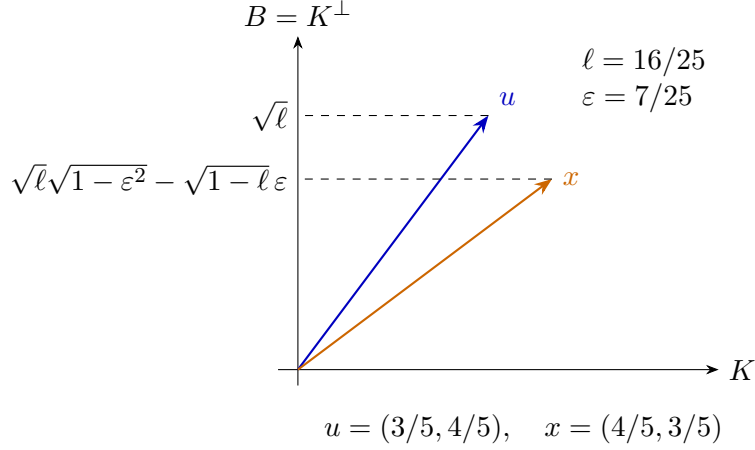


Figure 1: The sharp two-dimensional mechanism of UFR14 and IFE8-11. The plotted auxiliary example uses $u = (3/5, 4/5)$ and $x = (24/25)u + (7/25)(4/5, -3/5) = (20/25, 15/25)$. The figure coordinates and labels are specified exactly in the source; the generic formula, rather than the example, is applied to the original arithmetic subspaces.

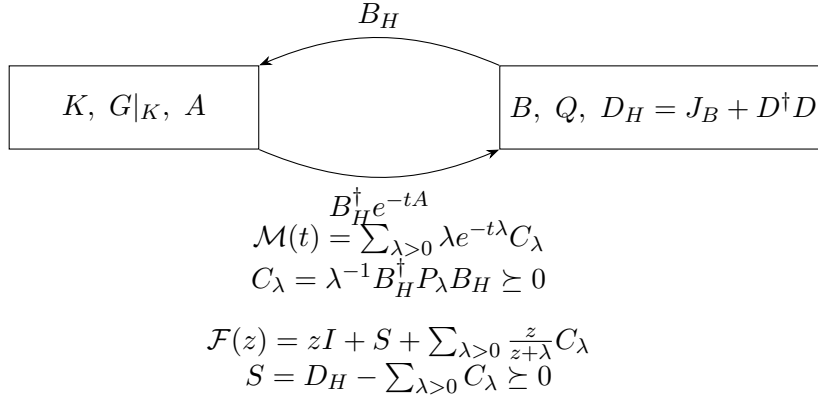


Figure 2: Exact domains and return maps in UH2-13. Arrows show the kernel-to-observation memory mechanism; box sizes do not represent dimensions. Each residue is the complete positive matrix. It may fail to commute with the other residues. The quotient map and its minimum section identify the resulting resolvent with $\mathcal{F}(z)^{-1}$.

Part I

Original-metric receiving proofs

1 Original source families: angle variation and sharp rational-filter detection

This derivation receives the complete supplied equations (1)–(87) without changing the original value map, physical coordinate, relation polynomial, source metric, or observation. It adds an exact angle bound for several source weights, extends the rational coefficient inverse to its full degree guard, and improves the retained slow-mode observation estimate.

1.1 1. A global angle estimate retaining every source group

Let $G_\circ > 0$ in the original fixed coefficient space E , and let the complete positive source covariances be $\Omega_i = F_i H_i^+ F_i^*$, with every cross term of each normal-source Gram H_i retained. Set

$$C(\mathbf{x}) = G_\circ^{-1} + \sum_{i=1}^p e^{x_i} \Omega_i, \quad G(\mathbf{x}) = C(\mathbf{x})^{-1}. \quad (\text{UFR1})$$

The prescribed physical source is its prescribed point in this family. The following identities do not assign unchanged physical norms to other points.

Fix disjoint subspaces $K, V \subset E$ with full frames I_K, U , dimensions m, r , and concatenated frame $[I_K, U]$ on $K + V$. Define

$$\delta(\mathbf{x}) = \log \det(I_K^* G I_K) + \log \det(U^* G U) - \log \det([I_K, U]^* G [I_K, U]). \quad (\text{UFR2})$$

Completing the full quadratic minimum over either subspace proves

$$\delta = \log \det H_K - \log \det \hat{H}_K = \log \det H_V - \log \det H_V^B \geq 0. \quad (\text{UFR3})$$

Here $\hat{H}_K$ is the actual quotient metric after removing V , and H_V^B is the actual metric after the observation with kernel K . These are the same Schur complements as IQR3 and IQR15–23 and CA6–11 in the accompanying complete derivations.

Take the exact straight path $\mathbf{x}(s) = \mathbf{x} + s\mathbf{v}$, $0 \leq s \leq 1$, with $\mathbf{v} = \mathbf{y} - \mathbf{x}$, and put

$$a = \min(0, v_1, \dots, v_p), \quad b = \max(0, v_1, \dots, v_p), \quad D_s = G(s)^{1/2} C'(s) G(s)^{1/2}. \quad (\text{UFR4})$$

The positive matrices

$$K_0(s) = G(s)^{1/2} G_\circ^{-1} G(s)^{1/2}, \quad K_i(s) = e^{x_i(s)} G(s)^{1/2} \Omega_i G(s)^{1/2}$$

sum to I . Consequently $aI \preceq D_s = \sum_i v_i K_i(s) \preceq bI$.

In the same exact isometry let P_K, P_V, P_{K+V} denote the Euclidean orthogonal projections, and set $A_s = P_K + P_V - P_{K+V}$. Differentiating all three original fixed-frame Grams, using $G' = -G C' G$, gives

$$\delta'(s) = -\text{Tr}(D_s A_s). \quad (\text{UFR5})$$

The complete principal-pair proof CA17–20 gives eigenvalues $+c_j, -c_j$, $0 \leq c_j < 1$, for A_s , with at most $\min(m, r)$ pairs; all other eigenvalues vanish. Thus $\text{Tr } A_s = 0$. Write $A_s = A_+ - A_-$, where $\text{Tr } A_+ = \text{Tr } A_- = \sum_j c_j$. The matrix $B_s = D_s - aI$ lies between 0 and

$(b-a)I$. Each of $\text{Tr}(B_s A_\pm)$ lies between zero and $(b-a) \sum_j c_j$. The absolute difference of these two numbers is at most that same upper bound. Integration proves

$$|\delta(\mathbf{y}) - \delta(\mathbf{x})| \leq \min(m, r) [\max(0, \mathbf{y} - \mathbf{x}) - \min(0, \mathbf{y} - \mathbf{x})]. \quad (\text{UFR6})$$

The maximum and minimum include every coordinate and zero. The proof keeps noncommuting source groups, changing metric angles, and all covariance eigenvalues. No bound on the covariance norm or its rank is required.

For the original $m = 8k - 16$, $q = (k+1)^2$, four separate endpoints with respective logarithmic weight changes of width Δ_N , this gives the explicit receiving estimate

$$|\mathcal{R}[\delta_N(\mathbf{y}_N) - \delta_N(\mathbf{x}_N)]| \leq \min(m, r) \sum_{N=q-1, q, 2q-1, 2q} \Delta_N. \quad (\text{UFR7})$$

In particular, changing each positive source weight by factors lying between q^{-A} and q^A , for fixed A , costs at most $8A \min(m, r) \log q = O(k \log q) = o(kq)$. This conclusion concerns the change from the actual starting source. Its initial angle remains present. Reaching a zero source weight requires a separate endpoint estimate; no logarithm of zero was inserted.

The individual restriction and quotient determinant losses have the exact difference

$$\overline{F}_K(\mathbf{x}, \mathbf{y}) - F_K(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{y}) - \delta(\mathbf{x}), \quad (\text{UFR8})$$

with each loss defined as initial minus final log determinant. Thus UFR7 directly updates their original four-cutoff comparison.

1.2 2. Full rational coefficient recovery at the sharp degree guard

Keep the physical coordinate S , complete upper and lower relations $\chi, \hat{\chi}$, and original shifts

$$\mathcal{T}_A f(S) = \sum_{j=1}^J a_j f(S + b_j), \quad E_A(z) = \sum_j a_j e^{b_j z}, \quad v = \text{ord}_0 E_A < \infty.$$

The complete lower quotient has degree $q' = \deg \hat{\chi}$, with $\hat{\chi}(S) \mid \chi(S + b_j)$. These are the retained conductor maps of IK1-13. The complete original observation satisfies $K = \ker \Lambda \subseteq \ker C_A$.

Let D be monic, $\deg D = d \geq 1$, $(D, \chi) = 1$, $d \leq q = \deg \chi$. Put $L = Jd$, $\mathcal{B} = \prod_j D(S + b_j)$, and $U_D p = [p/D]$, for $\deg p < d$. Coprimality and the divisibilities imply $(\mathcal{B}, \hat{\chi}) = 1$. For

$$N_p = \mathcal{B} \mathcal{T}_A(p/D)$$

the expansion at infinity gives

$$\deg N_p \leq L - v - 1. \quad (\text{UFR9})$$

To prove it, write a nonzero proper rational function as $f = cS^{-n_0} + O(S^{-n_0-1})$, $n_0 \geq 1$. The coefficient of S^{-n_0-v} in its translated sum is $(-1)^v \binom{n_0+v-1}{v} \mu_v c \neq 0$, because every earlier symbol moment vanishes. The same argument proves that $\mathcal{T}_A$ has no nonzero proper-rational kernel.

Let T_D be the coefficient matrix of N_{S^ℓ} , $0 \leq \ell < d$, in the full rows $0, \dots, L-1$. It has full column rank. Define $\text{coef}_{<L}$ on the lower quotient by its unique degree-below- q' remainder, truncating or padding with zeros as necessary. On the full guard $L - v \leq q'$, the numerator N_p is already that remainder. Therefore

$$W_D = (T_D^* T_D)^{-1} T_D^* \text{coef}_{<L} M_B^{(-)}, \quad \boxed{W_D C_A U_D = I_d}. \quad (\text{UFR10})$$

This proves the supplied formula (58) on the larger guard $Jd - v \leq q'$, including $q' < Jd$; its last v numerator coefficients are exactly zero. Nothing is inferred from values at only some of the lower roots. Full Hermite jets are converted by their original invertible Hermite map.

For families in which v varies, the uniform guard $Jd \leq q'$ remains available independently of v . On a fixed-order stratum the sharper guard is exact. The formula uses no inverse pole difference; polynomial numerator coordinates preserve the full collision.

Let $Q = (\Lambda G^{-1} \Lambda^*)^{-1}$, $\Gamma_A = C_A G^{-1} C_A^*$, $H_D = U_D^* G U_D$. Every lift of the observed value is also a lift of its conductor value, because $C_A = R_A \Lambda$. Minimizing over the larger conductor fibre proves

$$U_D^* \Lambda^* Q \Lambda U_D \succeq (C_A U_D)^* \Gamma_A^+ (C_A U_D). \quad (\text{UFR11})$$

The pseudoinverse acts on the actual conductor image. If C_A is onto, it is the usual inverse. This formulation also covers a smaller conductor image without deleting it. UFR10 implies $\|C_A U_D a\|_2 \geq \|a\|_2 / \|W_D\|$. On the positive range of Γ_A , its inverse is at least $1/\lambda_{\max}(\Gamma_A)$. Hence

$$U_D^* \Lambda^* Q \Lambda U_D \succeq \ell_D H_D, \quad \ell_D = \frac{1}{\lambda_{\max}(\Gamma_A) \|W_D\|^2 \lambda_{\max}(H_D)} > 0. \quad (\text{UFR12})$$

The left side is at most H_D , so the proved factor is at most one. Every matrix is in its specified original coefficient coordinates. The original unit, if included in the actual value map, remains in both maps and metrics.

The rational complexity bound of supplied (3) follows without another condition-number estimate:

$$x \in K \setminus \{0\} \implies d_{\text{rat}}(x) \geq \left\lfloor \frac{q' + v}{J} \right\rfloor + 1.$$

Indeed any smaller rational representative would satisfy UFR10, contradicting $C_A x = 0$. The complexity exists: $D = \chi + 1$ is coprime to χ , and the original remainder numerator has degree below q . This is a degree exclusion on the complete original kernel, uniform over all pole choices satisfying coprimality.

1.3 3. A sharper observed fraction for the entire filtered slow space

All norms and projections in this section use the same original G_N . Let $U \subset E$ be a subspace on which

$$\|P_B u\|^2 \geq \ell \|u\|^2, \quad 0 < \ell \leq 1. \quad (\text{UFR13})$$

Let V be a subspace satisfying $\|(I - P_U)P_V\| \leq \varepsilon$, $0 \leq \varepsilon \leq 1$. Then

$$\boxed{\|P_B x\|^2 \geq \left(\sqrt{\ell} \sqrt{1 - \varepsilon^2} - \sqrt{1 - \ell} \varepsilon \right)_+^2 \|x\|^2 \quad (x \in V).} \quad (\text{UFR14})$$

Here is a complete proof. UFR13 implies $\|P_K P_U\| \leq \sqrt{1 - \ell}$, where $K = B^\perp$; adjointing gives the same bound for $P_U P_K$. For unit $x \in V$, write $x = \sqrt{1 - t^2} u + t w$, with $u \in U$, $w \perp U$, both unit when their coefficient is nonzero, and $t \leq \varepsilon$. For a unit $z \in K$, put $a = \|P_U z\| \leq \sqrt{1 - \ell}$. The two orthogonal components give

$$|\langle z, x \rangle| \leq a \sqrt{1 - t^2} + \sqrt{1 - a^2} t.$$

If $\varepsilon < \sqrt{\ell}$, then $t < \sqrt{\ell}$ and $\sqrt{1 - t^2} > \sqrt{1 - \ell}$. The right side increases in a throughout the allowed interval, and its resulting value increases in $t \leq \varepsilon$. It is at most

$\sqrt{1-\ell}\sqrt{1-\varepsilon^2} + \sqrt{\ell}\varepsilon$. Taking the supremum over $z \in K$, and then using $\|P_B x\|^2 = 1 - \|P_K x\|^2$, proves UFR14 by expanding the square. If $\varepsilon \geq \sqrt{\ell}$, its claimed lower bound is zero, which proves the other case. Zero components follow by continuity. The argument also covers $\ell = 1$.

The factor in UFR14 improves supplied (70), which subtracts ε in place of $\sqrt{1-\ell}\varepsilon$. It becomes strictly positive exactly when $\varepsilon < \sqrt{\ell}$. The improvement is sharp: in a Euclidean two-plane take $K = \mathbb{C}e_1$, $U = \mathbb{C}(\sqrt{1-\ell}, \sqrt{\ell})$, and rotate its unit vector toward e_1 by $\arcsin \varepsilon$, when $\varepsilon < \sqrt{\ell}$. Its observed coordinate is exactly the positive expression in UFR14. This is an auxiliary sharpness example, not a choice of new arithmetic data.

Now retain a monic filter $p(y)$ of degree $1 \leq d \leq q$, coprime to Q_k , with $N \geq q-1$ and $d \leq N$. The last inequality includes every original cutoff unless $d = q, N = q-1$; that harmless endpoint is handled by a zero polynomial-quotient map and $B_p = 0$. Divide the complete original minimum polynomial section:

$$p(M)^{-1} = B_p + U_p T_p, \quad B_p = J_N \Omega_p \hat{R}_N, \quad T_p = \Re_p \hat{R}_N, \quad U_p r = [r/p]. \quad (\text{UFR15})$$

The equality follows by multiplying the exact polynomial division identity by $p(M)^{-1}$. Its first term has $\|B_p\| \leq D_p$, where D_p is the operator norm of the full native polynomial-division map; the minimum lift is an isometry, and the original value map is contractive. No rational function is integrated at a pole.

Let $X = p(M)^{-1}$, let $\sigma_j(X)$ be decreasing singular values, and let V_d be any top d left singular space. For $x = Xu \in V_d$, its right singular preimage satisfies $\|u\| \leq \|x\|/\sigma_d(X)$. Because the finite-rank part of UFR15 lies in U_p ,

$$\|(I - P_{U_p})x\| \leq D_p \|u\|, \quad \varepsilon_p = \min(1, D_p/\sigma_d(X)).$$

The rank- d term also gives $\sigma_{d+1}(X) \leq D_p$: on its nullspace of codimension at most d , $X = B_p$, and the variational characterization of singular values applies.

Use ℓ_p from UFR12 on the full conductor degree guard, and put

$$\ell_{\text{sharp}} = \left(\sqrt{\ell_p} \sqrt{1 - \varepsilon_p^2} - \sqrt{1 - \ell_p} \varepsilon_p \right)_+^2. \quad (\text{UFR16})$$

The singular-vector expansion, whose weights are the actual orthogonal projection norms, now proves at every $\tau \geq 0$

$$\begin{aligned} \text{Tr}(P_B e^{-\tau p(M)^\dagger p(M)}) &\geq \ell_{\text{sharp}} \sum_{j=1}^d e^{-\tau/\sigma_j(X)^2}, \\ \text{Tr}(P_K e^{-\tau p(M)^\dagger p(M)}) &\leq (1 - \ell_{\text{sharp}}) \sum_{j=1}^d e^{-\tau/\sigma_j(X)^2} + (q-d)e^{-\tau/D_p^2}. \end{aligned} \quad (\text{UFR17})$$

The last term is omitted when $d = q$; if $D_p = 0$, invertibility and UFR15 force $d = q$. Thus no division by zero is used.

Finally the exact small positive semidefinite matrix

$$\mathcal{S}_p = H_p^{1/2} T_p G_N^{-1} T_p^* H_p^{1/2}, \quad H_p = U_p^* G_N U_p \quad (\text{UFR18})$$

has as its eigenvalues the **squared** singular values of $U_p T_p$, with zeros included when its rank is below d . Indeed $U_p H_p^{-1/2}$ is an isometry; after this isometry the map is $H_p^{1/2} T_p$, whose product with its adjoint is $\mathcal{S}_p$. This corrects the unsquared wording following supplied (72). The strict guard below proves positive definiteness when it is used.

Let $\mu_d = \lambda_{\min}(\mathcal{S}_p)$. The singular-value variational principle and $\|B_p\| \leq D_p$ give

$$\sigma_d(X) \geq \sqrt{\mu_d} - D_p.$$

Therefore the explicitly evaluable sufficient finite guard

$$\boxed{\sqrt{\mu_d} > D_p(1 + \ell_p^{-1/2})} \quad (\text{UFR19})$$

makes UFR17 strictly positive with ε_p replaced by $D_p/(\sqrt{\mu_d} - D_p)$. All matrices in this guard come from the unchanged full source and observation. Without that inequality the finite bound remains valid and can be zero; no positivity is asserted from rank alone.

1.4 4. Exact physical coordinate and the joint arithmetic norm

The monic physical denominator corresponding to the monic filter $p(y)$ is

$$D(S) = i^d p((S - k/2)/i), \quad (Tr)(S) = i^d r((S - k/2)/i). \quad (\text{UFR20})$$

The map T is invertible on degree-below- d numerators. The original algebra identification $\Phi([f(y)]) = [f((S - k/2)/i)]$ obeys $\Phi U_p = U_D T$, because the identical factors i^d cancel in the rational function. The full relation ideals correspond under the same substitution, including every root multiplicity. Transporting the original metric by Φ , the source and observed Grams both change by congruence with T . Thus the very same scalar ℓ_D in UFR12 applies to U_p ; no additional physical-coordinate factor is discarded. Independent IFE30–31 records both maps.

For the original simple quartet, retain $k \equiv 1 \pmod{4}$, $k \geq 5$, $q = (k + 1)^2$, fixed $0 < \delta < 1/2$, $\gamma > 2$, and every allowed original period and cutoff. In the exact complete value coordinates,

$$y_{ab} = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k, \quad C_k = k\sqrt{\delta^2 + \gamma^2}. \quad (\text{UFR21})$$

The full tensor value metric $G_{k,1}$ is diagonal in these simple-root values, with all its positive ordered-tuple weights retained. Multiplication by y is therefore a diagonal operator of norm exactly C_k : its upper bound is attained at a corner (a, b) . This coordinate calculation is an isometry through the complete value map, not a replacement of the original coefficient metric.

The complete earlier source theorem J3, reproduced in full in this edition, proves before minimization

$$\Lambda_{h,2q}^{-k} G_{k,1} \preceq G_N^J \preceq \tau_\eta^k G_{k,1}, \quad W_k = k \log(\tau_\eta \Lambda_{h,2q}) = O_h(k \log^2(q + 2)).$$

Here the same tensor section and its full cross Gram are used at all four cutoffs; its physical tolerance is the fixed simple-quartet choice $\eta = 1$. Applying the two bounds to Mv and v proves $\|M\|_{G_N^J}^2 \leq C_k^2 e^{W_k}$. For any positive source-weight segment based at this actual joint metric, let $\Delta = \max(0, v_i) - \min(0, v_i)$, with v_i the logarithmic weight ratios. Comparing every positive covariance summand gives the two form bounds with ratio e^Δ ; taking inverses and then the same Rayleigh quotient proves

$$\|M\|_{G_N(\mathbf{x})}^2 \leq C_k^2 e^{W_k + \Delta}. \quad (\text{UFR22})$$

All covariance groups here are the prescribed full normal-source groups from that joint metric. This estimate does not transfer the joint-source enclosure to the unactivated scalar metric.

Apply the complete-memory determinant identity UH29-31 at positive regularizer z . If $F_N(z)$ is the original observed resolvent pencil and $M_{B,N} = \Lambda M L_N$, those finite proofs give

$$\left| \log \det F_N(z) - \log \det(zI + M_{B,N}^\dagger M_{B,N}) \right| \leq (8k - 16) \log \left(1 + \frac{C_k^2 e^{W_k + \Delta}}{z} \right). \quad (\text{UFR23})$$

Indeed the leakage and memory ranks are at most $8k - 16$, while both squared-action block norms are at most the full action norm squared. This proves the displayed bound without discarding either correction. If the four source-weight ratios are between q^{-A} and q^A , and $z \geq e^{-\eta_k q}$ with $\eta_k \rightarrow 0$, their widths obey $\Delta \leq 2A \log q$. For all sufficiently large k , the logarithm is at most $\log 2 + 2 \log C_k + W_k + 2A \log q + \eta_k q = o(q)$. Adding the four exact absolute bounds proves an $o(kq)$ error for the original four-cutoff determinant comparison throughout this whole regularizer window. The estimate includes the full observed memory pencil; the canonical scalar kernel determinant and complex-current phases remain their separate, explicitly retained receivers.

1.5 5. Relation to the preceding kernel calculation

UFR6-8 control how the exact quotient angle varies across positive source weights even when the source covariance has full rank. IQR22-23 separately evaluates the scalar adjacent-source update using the retained monic relation norm. These statements improve the kernel receiving estimates while preserving the unevaluated initial determinant.

The complete TC1-29 proof in this edition now evaluates the actual ACC24 two-column update as well. Its exact scalar and residual joint-source contributions obey

$$0 \leq D_N(\eta) \leq \log \frac{\nu_{N+1-q}}{\omega_{N+1}} + \log(\mathbf{BL}) = O_h(q), \quad 0 \leq \eta \leq 1. \quad (\text{UFR24})$$

TC7-8 retains the full complex cross term, and TC10-12 derives the second contribution's rank-one bound from the same full-source enclosure, without a dimension multiplier. TC20-22 consequently proves an $o(kq)$ angle change from the actual activated baseline at every original cutoff. This strengthens the finite rank-two receiving estimate in IQR15-19. The exact relation to the full normal-source covariance is $\Omega_{N+1} - \Omega_N = g_N g_N^* - f_N f_N^*$, as proved in TC24; the full normal-source determinant is retained in TC25.

UFR9-12 extend the existing inverse-power conductor map to all proper rational numerators below the full degree guard, including colliding poles. UFR14-19 then transfer its actual metric lower bound to a whole slow singular space of a polynomial filter. The filter is $p(M)$ on the original quotient; it is never replaced by $p(\Lambda M L_N)$.

The independent heat derivation accompanying this note proves the complete supplied memory and leakage bounds. In particular it compares the two different positive heat traces with error at most $8k - 16$, uniformly in time. That controls their q -scale comparison. Their complex projected-current phases still require their original coefficients.

The original-author sources retained with this programme include the Stacks Project Authors' cochain homotopy definition and quasi-isomorphism definition, for the exact observation-complex interpretation in the preceding quotient proof. Their original TeX is included; these definitions are not the authorship of the finite inequalities derived here. The arithmetic polynomial norms retain T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw's NIST DLMF Chapter18 conventions, with W. P. Reinhardt: 18.22.E8 and 18.23.E7. No external theorem is invoked in place of the complete finite proofs above.

Part II

The actual activated two-column source

2 The actual activated two-column update

21 September 2026. This is an independent finite derivation of the adjacent update ACC24, retaining its activated baseline, complete physical source, and every value cross term. It also identifies exactly which distinct two-column update admits a product of consecutive monic-norm ratios.

The actual ACC24 update is one scalar outgoing column together with one outgoing column of the full joint source. Its determinant is not the determinant of two consecutive scalar polynomial additions. Nevertheless its logarithm is proved below to be $O_h(q)$, uniformly over $0 \leq \eta \leq 1$, at each of the four original cutoffs. The joint part has the stronger bound $O_h(k \log^2(q+2)) = o(q)$. Hence the complete angle change is $o(kq)$ from the actual activated baseline, with no assertion about its unevaluated initial angle.

2.1 Source identities and exact reading coverage

All paths in this paragraph are relative to the `spectral_intake_20260921` directory containing this note's parent directory. The sources read for this calculation were:

- `inverse_sector_reduction_20260921/INVERSE_QUOTIENT_RECEIVERS.tex`, IQR15-23, SHA-256 251B10623FB60C425775E62B434C744F3C2F14CF00EE2D334B63B0E2D0B85CAC.
- `inverse_sector_reduction_20260921/COMPLETE_MIXED_SOURCE_PROOFS.tex`, C6-10, lines 523-552, SHA-256 82CCCE2D56D43C544B1D813BDA27DBE1C20700FAA83BD983E34C3D83315C0C99. The surrounding current/source discussion was also read. Its actual public locator is C6-10 in the complete mixed-source proof.
- `ORIGINAL_ADJACENT_CUTOFF_CURRENT.tex`, complete file; especially ACC24 at line 324 and ACC27 at line 360. SHA-256 786933325827D18FFD924C73C218AA1CED9CB4F0C232830A814488825A5507CB.
- `activation_intake_20260921/ACTIVATION_ORIGINAL_COMPLETE_PROOFS.md`, opening definitions and A1-19, the finite return proof through A19, and A28-31. Exact locators: A4 line 62, A7 line 108, A9 line 127, A28 line 522, A29 line 532, A30 line 540. SHA-256 6FC047F2513C6C61E0568E47BDA23CCB5CCC4394E7704BB7F3F4E5AA9D871E29.
- `../JOINT_MINIMUM_COMPLETE.tex`, lines 611-683: J3 is proved at lines 634-643, and the complete normal-source maps J5-J8 occur at lines 655-671. SHA-256 70DC576A470E1FDC15AFE7C7CA49EFCDD569BF1EA2A429094D0E19B97EBCD416.
- `publication_003/RETAINED_PROFILE_PROOF.tex`, opening source definitions and H1-9; H3 is the exact monic norm calculation and H4 is the retained profile through relation degree $q+1$. Its public locator is H3-H4 in the retained original profile.

The exact public proof sources are IQR15-23, ACC24-27, A6-9 and A28-30 in the complete activation reader, and the complete J3 source included in this edition.

These are bounded readings of the named original sources, not an exhaustive literature claim. The finite identities below are proved here; the named original coercivity bounds and monic profile are retained with their stated source scope. Their arithmetic source conventions retain T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, NIST DLMF 18.22.E8 and 18.23.E7. The one-factor analytic bounds retain T. M. Apostol's Chapter 25 and R. A. Askey and R. Roy's Chapter 5 attributions in the full J3 source and its accompanying original equation TeX.

2.2 1. Original spaces and the actual two columns

Keep the original physical coordinate $S = k/2 + iy$, full monic relation $Q_k(y) = i^{-q} \chi_k(k/2 + iy)$, and source measure $d\mu = w_h^{*k}(y) dy$. Write $E = \mathbb{C}[y]/(Q_k)$ in its unchanged coefficient coordinates. Let X_N be the original scalar polynomial source through degree N , represented in the original physical Hilbert space. Let Y be the same fixed tensor-section image at every cutoff, and write $\mathcal{J}_N = X_N + Y$.

Every attained metric uses the full prescribed tensor value, including the original unit and the complete relation kernel. Let

$$C_N = G_N^{-1}, \quad C_N^J = (G_N^J)^{-1}, \quad C_N(\eta) = (1 - \eta)C_N + \eta C_N^J, \quad G_N(\eta) = C_N(\eta)^{-1}, \quad (\text{TC1})$$

for source activation $0 \leq \eta \leq 1$. The tensor-section parameter, when it must be named, will be ζ ; it is fixed and is not the activation η .

Set $n = N + 1$, and let $\phi_n = p_n/\sqrt{\omega_n}$ be the actual orthonormal outgoing scalar polynomial. Define

$$f_N = \text{value}(\phi_n), \quad z_N = (I - P_{\mathcal{J}_N})\phi_n, \quad g_N = \begin{cases} \text{value}(z_N)/\|z_N\|, & z_N \neq 0, \\ 0, & z_N = 0. \end{cases} \quad (\text{TC2})$$

The projections here are physical source projections. A nonzero z_N may still have zero value, in which case $g_N = 0$ as well. The exact orthogonal source decompositions are

$$X_{N+1} = X_N \perp \mathbb{C}\phi_n, \quad \mathcal{J}_{N+1} = \mathcal{J}_N \perp \mathbb{C}z_N, \quad (\text{TC3})$$

with the last summand absent if $z_N = 0$. Covariances add under these actual orthogonal decompositions, so

$$\begin{aligned} C_{N+1} &= C_N + f_N f_N^*, & C_{N+1}^J &= C_N^J + g_N g_N^*, \\ C_{N+1}(\eta) &= C_N(\eta) + U_N U_N^*, & U_N &= [\sqrt{1 - \eta} f_N, \sqrt{\eta} g_N]. \end{aligned} \quad (\text{TC4})$$

This proves the exact ACC24 update from its source spaces. Zero or dependent value columns remain allowed.

The two coefficient channels in U_N belong to the two weighted covariance summands in TC1. They are not asserted to be orthogonal vectors in their common physical ambient source. Indeed put $Z_N = (I - P_{X_N})Y$, so $\mathcal{J}_N = X_N \perp Z_N$, and use $\phi_n \perp X_N$. Then

$$z_N = (I - P_{Z_N})\phi_n, \quad f_N = h_N + \sqrt{\theta_N} g_N, \quad h_N = \text{value}(P_{Z_N}\phi_n), \quad \theta_N = \|z_N\|^2 \in [0, 1]. \quad (\text{TC5})$$

The formula holds also at $\theta_N = 0$. It is the exact connecting value identity. In particular the second column is a residual of the same degree- n polynomial against the old full joint source; it is not $[\phi_{n+1}]$.

For completeness, TC1 is the covariance of the value map

$$(x, j) \mapsto \sqrt{1-\eta} \text{value}(x) + \sqrt{\eta} \text{value}(j)$$

on the orthogonal direct sum $X_N \oplus \mathcal{J}_N$. After splitting $\mathcal{J}_N = X_N \perp Z_N$, rotate the two copies of X_N by the unitary two-by-two coefficient matrix with rows $(\sqrt{1-\eta}, \sqrt{\eta})$ and $(-\sqrt{\eta}, \sqrt{1-\eta})$. The first copy supplies the unchanged scalar value map and the second has zero value. Removing only that zero-value channel leaves $X_N \oplus Z_N$ with the prescribed normal-source penalty η^{-1} . This proves the exact relation to A11 without deleting a cross term of an unprocessed source Gram.

2.3 2. The full finite determinant with all cross terms

Fix N, η , abbreviate $G = G_N(\eta)$, $f = f_N$, $g = g_N$, and define

$$a = (1-\eta)f^*Gf, \quad c = \eta g^*Gg, \quad b = \sqrt{\eta(1-\eta)}f^*Gg. \quad (\text{TC6})$$

The determinant identity $\det(I+AB) = \det(I+BA)$, proved by eliminating either block of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$, gives

$$e^{D_N(\eta)} := \frac{\det C_{N+1}(\eta)}{\det C_N(\eta)} = \det(I_2 + U_N^* G U_N) = (1+a)(1+c) - |b|^2. \quad (\text{TC7})$$

Equivalently, if $G_1 = (G^{-1} + (1-\eta)f f^*)^{-1}$, then direct multiplication gives

$$G_1 = G - \frac{1-\eta}{1+a} G f f^* G, \quad e^{D_N(\eta)} = (1+a)(1+\eta g^* G_1 g), \quad (\text{TC8})$$

because $\eta g^* G_1 g = c - |b|^2/(1+a)$. Thus the interference term is retained exactly in the conditional second factor.

For any other independent frame on these same two source channels, write $F = U_N T$, $H = T^* T$, with T invertible. Its complete source covariance satisfies

$$F H^{-1} F^* = U_N U_N^*, \quad \det(I + G^{1/2} F H^{-1} F^* G^{1/2}) = \frac{\det(H + F^* G F)}{\det H} = e^{D_N(\eta)}. \quad (\text{TC9})$$

No diagonal replacement of H occurs. If a redundant source description is used, pass to its positive source range with the original kernel compatibility, or retain its Moore-Penrose inverse; its nonzero covariance eigenvalues and the determinant in TC9 are unchanged.

2.4 3. A rank-one joint step has no dimension multiplier

The full-source enclosure needed here is the actual J3/A7 enclosure in one common physical value frame:

$$\mathbf{L}^{-1} T \preceq G_M^J \preceq \mathbf{B} T, \quad T = G_{k,1}, \quad (\text{TC10})$$

at both $M = N$ and $M = N + 1$. It is obtained before minimization on the lower side, and by the retained full tensor section on the upper side. More explicitly, J3 gives $\Lambda_{h,n_*}^{-k} T \preceq G_M^J \preceq \tau_\zeta^k G_{k,\zeta} \preceq \tau_\zeta^k T$ after choosing the common degree allowance to cover both cutoffs and the fixed section. Thus one may take $\mathbf{L} = \Lambda_{h,n_*}^k$, $\mathbf{B} = \tau_\zeta^k$, or the

already retained A7 constants. Here the existing $n_* = 2q + 2$ allowance covers every outgoing degree used below; if the fixed section degree is larger, use its maximum with $2q + 2$, as required by J3. In the original eventual domain the fixed section degree is already below this allowance.

TC10 implies $\mathbf{BL} \geq 1$. Inverting its lower bound at $N + 1$, and applying its upper bound at N , yields

$$\begin{aligned} I + (G_N^J)^{1/2} g_N g_N^* (G_N^J)^{1/2} &= (G_N^J)^{1/2} (G_{N+1}^J)^{-1} (G_N^J)^{1/2} \\ &\preceq \mathbf{L} (G_N^J)^{1/2} T^{-1} (G_N^J)^{1/2} \preceq \mathbf{BL} I. \end{aligned} \quad (\text{TC11})$$

For the last order, $G_N^J \preceq \mathbf{BT}$ is equivalent to $T^{-1/2} G_N^J T^{-1/2} \preceq \mathbf{BI}$. The positive matrix $(G_N^J)^{1/2} T^{-1} (G_N^J)^{1/2}$ has the same eigenvalues, hence the same upper scalar bound. This verifies the order without assuming the two metrics commute.

The left side of TC11 is an identity plus one rank-one positive matrix. Its only possibly nonunit eigenvalue is $1 + g_N^* G_N^J g_N$. Therefore

$$0 \leq D_N^J := \log \frac{\det C_{N+1}^J}{\det C_N^J} = \log(1 + g_N^* G_N^J g_N) \leq \log(\mathbf{BL}). \quad (\text{TC12})$$

This is the complete rank-one derivation. A factor q is unnecessary because all other relative eigenvalues equal one. It includes $g_N = 0$.

The original A6–7 bounds give

$$\log(\mathbf{BL}) = O_h(k \log^2(q + 2)) = o(q) \quad (\text{TC13})$$

at fixed original packet and fixed section data. In the simple-quartet application $q = (k + 1)^2$, so the last equality follows directly from $\log^2(q + 2)/k \rightarrow 0$. No new lower bound on an individual normal-source Gram eigenvalue is used.

2.5 4. Evaluated bound for the actual activated determinant

Because $X_N \subset \mathcal{J}_N$, the same-fibre minimum gives $G_N^J \preceq G_N$, hence $C_N^J \succeq C_N$. Thus

$$G_N(\eta) \preceq G_N, \quad a \leq \alpha_N := f_N^* G_N f_N. \quad (\text{TC14})$$

Also $G_1 \preceq G_N(\eta)$. For $\eta > 0$, TC1 gives $G_N(\eta) \preceq \eta^{-1} G_N^J$, so $\eta g_N^* G_1 g_N \leq g_N^* G_N^J g_N$. At $\eta = 0$, the former term is zero. Taking logarithms in TC8 proves the exact refinement of A30

$$0 \leq D_N(\eta) - \log(1 + (1 - \eta) \|f_N\|_{G_N(\eta)}^2) \leq D_N^J \leq \log(\mathbf{BL}). \quad (\text{TC15})$$

The scalar endpoint has an exact monic-norm value. Let $s = N + 1 - q$, and let ν_s be the squared minimum norm of a monic degree- $N + 1$ polynomial divisible by the complete Q_k . Its unique minimizer $Q_k v_s$ is orthogonal to all previous relations. The difference $Q_k v_s - p_{N+1}$ belongs to the old scalar space X_N , has value $-[p_{N+1}]$, and is orthogonal to every old zero-value relation. It is therefore the exact minimum old-source lift of that value. Since $p_{N+1} \perp X_N$,

$$\nu_s = \omega_{N+1} + \|Q_k v_s - p_{N+1}\|_\mu^2, \quad 1 + \alpha_N = R_N := \frac{\nu_s}{\omega_{N+1}}. \quad (\text{TC16})$$

Combining TC12–16 gives the uniform finite estimate

$$0 \leq D_N(\eta) \leq \log \frac{\nu_{N+1-q}}{\omega_{N+1}} + \log(\mathbf{BL}), \quad 0 \leq \eta \leq 1. \quad (\text{TC17})$$

Every source cross term remains in the exact TC7-8 identity; TC17 bounds that identity rather than replacing it.

At $N = q - 1, q, 2q - 1, 2q$, H4/IQR22 supplies

$$\log R_N = 2q\psi((N + 1 - q)/q) + O_h(k \log^2(q + 2)) = O_h(q), \quad (\text{TC18})$$

including its retained relation degree $q + 1$ at $N = 2q$. Hence TC17 is $O_h(q)$, uniformly in η , at all four actual cutoffs. This evaluates the needed determinant size at the actual activated baseline.

The $O_h(q)$ bound itself also follows from the complete finite coefficient bounds in TC29 below, so it does not require the sharper H4/EIQ asymptotic. TC18 and the fixed- b refinement TC19 retain that earlier profile theorem's stated proof status and parameter scope.

The already proved scalar profile A4 also evaluates the leading term when its source activation is $\eta_k(b) = e^{-2bq \log(q/k)}$, with fixed $b > 0$. Its stated result is $\log \|f_N\|_{G_N(\eta_k(b))} = q\Psi_b((N + 1 - q)/q) + o_h(q)$, uniformly in the original window, with positive profile there. Since $\eta_k(b) \rightarrow 0$, TC15 gives

$$D_N(\eta_k(b)) = 2q\Psi_b((N + 1 - q)/q) + o_h(q). \quad (\text{TC19})$$

On fixed compact subsets of $0 < b < 1$, the retained scalar error and TC13 give the explicit error $O_h(q/\log(q/k) + k \log^2(q + 2))$. At $\eta = 1$, TC12 gives instead $D_N(1) = D_N^J = o(q)$. No moving- b conclusion is obtained by substituting into a fixed- b asymptotic.

2.6 5. The resulting exact angle receiver

Fix the original disjoint K, V_r and their original coefficient frames at each cutoff. Let $\delta_{N,\eta}(t)$ be their concatenated-frame angle loss in $(C_N(\eta) + tU_N U_N^*)^{-1}$, for $0 \leq t \leq 1$. The previously independently proved IQR17 spectral-sum inequality gives

$$\begin{aligned} |\delta_{N,\eta}(t) - \delta_{N,\eta}(0)| &\leq \sum_{j=1}^{\min(m,r,2)} \log(1 + t\lambda_j(U_N^* G_N(\eta) U_N)) \\ &\leq \log \det(I_2 + U_N^* G_N(\eta) U_N) = D_N(\eta). \end{aligned} \quad (\text{TC20})$$

Zero eigenvalues contribute zero. Combining TC17-18 yields

$$|\delta_{N,\eta}(1) - \delta_{N,\eta}(0)| = O_h(q) = o(kq) \quad (\text{TC21})$$

uniformly in the actual activation η . For four possibly different activations η_N , keeping the four original signs,

$$|\mathcal{R}_N[\delta_{N,\eta_N}(1) - \delta_{N,\eta_N}(0)]| \leq \sum_{N=q-1,q,2q-1,2q} [\log R_N + \log(\mathbf{BL})] = O_h(q) = o(kq). \quad (\text{TC22})$$

The initial angle at $G_N(\eta_N)$ remains in these equations. Rank has not been used to replace it by the canonical initial angle. Through IQR19, TC22 controls the difference between the actual kernel determinant loss and its quotient loss, with both taken from this same activated baseline.

2.7 6. What the complete normal source actually evaluates

Let the actual normal source at cutoff N have its complete original Gram $H_{Z,N}$ and value map $F_{Z,N}$, as in J5. With the original positive-range inverse set

$$\Omega_N = F_{Z,N} H_{Z,N}^+ F_{Z,N}^*, \quad C_N^J = C_N + \Omega_N, \quad Z_N(\eta) = \det(I + \eta G_N^{1/2} \Omega_N G_N^{1/2}). \quad (\text{TC23})$$

These are the full normal-source covariances of J7-8 and C6-10; they can have rank of order q . They are not the rank-two covariance $U_N U_N^*$. Subtracting the two exact covariance increments in TC4 proves the connecting identity

$$\boxed{\Omega_{N+1} - \Omega_N = g_N g_N^* - f_N f_N^*} \quad (\text{TC24})$$

In particular the difference of two positive complete normal-source covariances need not be positive. Using $C_N(\eta) = C_N + \eta \Omega_N$ and the determinant lemma gives the exact factorization

$$\boxed{e^{\mathcal{D}_N(\eta)} = R_N \frac{Z_{N+1}(\eta)}{Z_N(\eta)}} \quad (\text{TC25})$$

Thus one canonical monic-norm ratio is present, together with the exact change of the full normal-source determinant. That second quotient contains the fixed tensor section and its complete scalar-section cross Gram. It cannot be replaced by the next canonical monic ratio merely because TC4 has two columns. TC25 is the exact relation to the full mixed-source response C6-10. For example, their logarithmic response identity reads $\eta \partial_\eta \log Z_N(\eta) = \text{Tr}[\eta G_N^{1/2} \Omega_N G_N^{1/2} (I + \eta G_N^{1/2} \Omega_N G_N^{1/2})^{-1}]$; differentiating TC25 connects the difference of those responses to the adjacent determinant, without identifying covariance eigenvalues with arithmetic ones.

TC15-19 evaluate the size, and in the stated fixed-activation regimes the leading term, of the complete left side of TC25. They do so without requiring an independent evaluation of both large normal-source determinants on its right side.

2.8 7. The different two-polynomial update and its exact monic product

For comparison within the same original polynomial source, adding the two consecutive orthonormal columns ϕ_{N+1}, ϕ_{N+2} to X_N gives a different covariance update

$$C_{N+2} = C_N + F_N F_N^*, \quad F_N = [\text{value}(\phi_{N+1}), \text{value}(\phi_{N+2})]. \quad (\text{TC26})$$

Applying TC16 successively with the actual updated baseline proves

$$\boxed{\det(I_2 + F_N^* G_N F_N) = \frac{\det C_{N+2}}{\det C_N} = \frac{\nu_{N+1-q} \nu_{N+2-q}}{\omega_{N+1} \omega_{N+2}}} \quad (\text{TC27})$$

The two columns can have a nonzero value cross Gram $f_{N+1}^* G_N f_{N+2}$. It is exactly included in the two-by-two determinant on the left. Sequential evaluation accounts for it by measuring the second column in G_{N+1} , not again in G_N . The invariance formula TC9 proves the same identity for every full non-diagonal source Gram obtained by changing the two normal-source coordinates.

At an already activated baseline $G_N(\eta)$, adding those same two scalar columns yields $\det(I_2 + F_N^* G_N(\eta) F_N)$, which need not equal TC27. The exact connecting formula is

$$\det(I_2 + F_N^* G_N(\eta) F_N) = R_N R_{N+1} \frac{\det(I + \eta G_{N+2}^{1/2} \Omega_N G_{N+2}^{1/2})}{\det(I + \eta G_N^{1/2} \Omega_N G_N^{1/2})}. \quad (\text{TC28})$$

Here Ω_N is held fixed because this is the two-scalar-column update, not the next full joint source. TC28 follows directly by evaluating $\det(C_{N+2} + \eta \Omega_N) / \det(C_N + \eta \Omega_N)$ in the two orders. Since $G_N(\eta) \preceq G_N$, its logarithm is bounded above by $\log R_N + \log R_{N+1}$.

For a finite $O_h(q)$ bound that also covers $N = 2q$ in TC27, one must retain the extra degree $N + 2 = 2q + 2$. The quoted H4 profile is stated through relation degree $q + 1$, so it is not used beyond that range. The existing full polynomial coefficient bounds A9, through $n_* = 2q + 2$, suffice. If all roots of the full Q_k have modulus at most R , they give for every $q \leq n \leq 2q + 2$

$$1 \leq \frac{\nu_{n-q}}{\omega_n} \leq \frac{g_+}{g_-} (1 + R/q)^{2q}. \quad (\text{TC29})$$

Indeed any degree- n monic polynomial has scaled leading coefficient q^n , so its squared source norm is at least $g_- q^{2n}$. The complete relation trial $Q_k(y) y^{n-q}$ is monic and admissible for ν_{n-q} . Its scaled coefficient one-norm is at most $q^n (1 + R/q)^q$, so its squared source norm is at most $g_+ q^{2n} (1 + R/q)^{2q}$. Taking the two minima proves TC29 with all roots and their multiplicities retained. The lower ratio follows because the relation-restricted monic minimum is taken over a subset of all monic polynomials.

Finally A9 gives $\log(g_+/g_-) = O_h(q + k \log^2(q + 2)) = O_h(q)$, while $R = O_h(k)$. Thus each logarithm in TC29 is $O_h(q)$, and both TC27 and its activated-baseline upper bound have $O_h(q)$ logarithm at all four cutoffs. This is a proof for the genuine two-degree polynomial update; it has not been substituted for ACC24.

2.9 Scope of the new conclusion

The finite identities retain the fixed packet, complete arithmetic unit, original physical value map, full relation polynomial, same section Y at every cutoff, and its entire cross Gram. The $O_h(q)$ and $o(kq)$ conclusions use the existing simple-quartet domain $0 < \delta < 1/2$, $\gamma > 2$, $k \equiv 1 \pmod{4}$, $q = (k + 1)^2$, the original period and inverse-sector positivity guards, and the fixed-source bounds TC10/TC13 through the outgoing degree. The observation kernel keeps dimension $8k - 16$ on its proved five-orbit domain.

The actual ACC24 determinant is evaluated by TC7-8 and TC25, with the uniform bound TC17 and fixed-activation profile TC19. The corresponding original-metric angle change is TC20-22. The different consecutive-polynomial determinant has the exact monic product TC27. Each statement keeps its actual starting metric and does not evaluate an initial angle from a rank count.

Part III

Independent source-family and rational-filter proofs

3 Independent derivation of the source-family and filter receivers

21 September 2026. This derivation checks UFR6, UFR9-12, and UFR14-19 in ORIGINAL_FAMILY_RECEIVERS.md. The complete source file was read; its recorded SHA-256 is CB66F25C1DB1606FBEABB8F804DC75DB2EBBAE73C4CE0F36907221140487C395. The work below is an independent finite-dimensional derivation. It preserves each original source metric, quotient, physical coordinate, observation, and filter. It does not evaluate an unproved arithmetic positivity guard.

The estimates are correct with one correction to UFR18: its small matrix is positive semidefinite before the guard is checked, and its eigenvalues include zeros when the remainder map has deficient rank. UFR19 itself forces positive definiteness, so its conclusion is unchanged. The rational degree guard is sharp for the stated unreduced-numerator recovery; this does not assert failure of every possible conductor inverse beyond that guard.

3.1 1. Several logarithmic source weights

Let $G_\circ > 0$ be the original fixed coefficient metric, and let $\Omega_i \succeq 0$ be the complete prescribed source covariances, including all entries of each $F_i H_i^+ F_i^*$. Set

$$C(\mathbf{x}) = G_\circ^{-1} + \sum_{i=1}^p e^{x_i} \Omega_i, \quad G(\mathbf{x}) = C(\mathbf{x})^{-1}. \quad (\text{IFE1})$$

Fix disjoint subspaces K, V , of dimensions m, r , and fixed full frames I_K, U . Define the angle loss using exactly the concatenated frame:

$$\delta = \log \det(I_K^* G I_K) + \log \det(U^* G U) - \log \det([I_K, U]^* G [I_K, U]). \quad (\text{IFE2})$$

For a segment $\mathbf{x}(s) = \mathbf{x} + s\mathbf{v}$, $\mathbf{v} = \mathbf{y} - \mathbf{x}$, define $a = \min(0, v_1, \dots, v_p)$, $b = \max(0, v_1, \dots, v_p)$. The positive operators

$$K_0 = G^{1/2} G_\circ^{-1} G^{1/2}, \quad K_i = e^{x_i(s)} G^{1/2} \Omega_i G^{1/2}$$

sum to $G^{1/2} C G^{1/2} = I$. Thus

$$D = G^{1/2} C' G^{1/2} = \sum_i v_i K_i, \quad aI \preceq D \preceq bI. \quad (\text{IFE3})$$

Let $\tilde{P}_X$ be the Euclidean orthogonal projection onto $G^{1/2}X$, and set $A = \tilde{P}_K + \tilde{P}_V - \tilde{P}_{K+V}$. For any fixed frame I_X , differentiation of $G = C^{-1}$ gives

$$\frac{d}{ds} \log \det(I_X^* G I_X) = -\text{Tr}(D \tilde{P}_X), \quad \tilde{P}_X = G^{1/2} I_X (I_X^* G I_X)^{-1} I_X^* G^{1/2}. \quad (\text{IFE4})$$

Therefore $\delta' = -\text{Tr}(DA)$.

Here is the required spectrum without an assumption of orthogonality between the physical subspaces. Choose orthonormal frames of $G^{1/2}K$ and $G^{1/2}V$, and perform a singular-value decomposition of their cross Gram. Their paired unit vectors

satisfy $e_i^* f_j = c_i \delta_{ij}$, with $0 \leq c_i < 1$; equality $c_i = 1$ would contradict disjointness. In the orthonormal pair $e_i, (f_i - c_i e_i)/\sqrt{1 - c_i^2}$, the matrix of A is

$$\begin{pmatrix} c_i^2 & c_i \sqrt{1 - c_i^2} \\ c_i \sqrt{1 - c_i^2} & -c_i^2 \end{pmatrix}. \quad (\text{IFE5})$$

Its eigenvalues are $c_i, -c_i$. All unpaired directions and all directions outside the two-subspace sum give zero. Thus, writing $A = A_+ - A_-$,

$$\text{Tr } A = 0, \quad \text{Tr } A_+ = \text{Tr } A_- = \sum_i c_i \leq \min(m, r). \quad (\text{IFE6})$$

By IFE3, $0 \preceq D - aI \preceq (b - a)I$. Each number $\text{Tr}((D - aI)A_\pm)$ lies between zero and $(b - a) \sum_i c_i$. Their difference therefore has absolute value at most this latter bound. Since $\text{Tr } A = 0$, integration over $0 \leq s \leq 1$ proves

$$|\delta(\mathbf{y}) - \delta(\mathbf{x})| \leq \min(m, r) [\max(0, y_1 - x_1, \dots, y_p - x_p) - \min(0, y_1 - x_1, \dots, y_p - x_p)]. \quad (\text{IFE7})$$

This is UFR6. Neither commutation of the Ω_i nor a norm or rank estimate for them was used. The fixed source term in IFE1 accounts for the zero included in both extrema. All weights on the segment are positive; zero weights require a separate endpoint argument.

With the original four signs $+, +, -, -$, the triangle inequality gives UFR7. If each weight ratio belongs to $[q^{-A}, q^A]$, each width is at most $2A \log q$, hence the four-cutoff cost is at most $8A \min(m, r) \log q$. For $m = 8k - 16$, $q = (k + 1)^2$, this is $o(kq)$. Subtracting the exact Schur identity $\log \det \hat{H}_K = \log \det H_K - \delta$ at the two endpoints gives UFR8 with the initial-minus-final cost convention. No initial angle has been set equal to zero.

3.2 2. The optimal projection fraction

All operators in this section are orthogonal projections for the same original metric G_N . Let $P_B + P_K = I$, and suppose

$$\|P_B u\|^2 \geq \ell \|u\|^2 \quad (u \in U), \quad 0 < \ell \leq 1, \quad \|(I - P_U)P_V\| \leq \varepsilon \leq 1. \quad (\text{IFE8})$$

The first condition gives $\|P_K P_U\| \leq \sqrt{1 - \ell}$. Taking adjoints yields the identical bound for $P_U P_K$. For a unit $x \in V$, put $t = \|(I - P_U)x\| \leq \varepsilon$, so $x = \sqrt{1 - t^2}u + tw$, where $u \in U$, $w \perp U$ are unit whenever their coefficients are nonzero. For any unit $z \in K$, let $a_z = \|P_U z\| \leq \sqrt{1 - \ell}$. Orthogonal decomposition and Cauchy-Schwarz give

$$|\langle z, x \rangle| \leq a_z \sqrt{1 - t^2} + \sqrt{1 - a_z^2} t. \quad (\text{IFE9})$$

If $\varepsilon < \sqrt{\ell}$, the right side increases in a_z for $0 \leq a_z \leq \sqrt{1 - \ell}$: its derivative is nonnegative exactly when $a_z^2 + t^2 \leq 1$, which holds. After setting $a_z = \sqrt{1 - \ell}$, its value increases in $t \leq \sqrt{\ell}$, by the same derivative calculation. Thus

$$\|P_K x\| \leq \sqrt{1 - \ell} \sqrt{1 - \varepsilon^2} + \sqrt{\ell} \varepsilon.$$

The identity

$$1 - \left(\sqrt{1 - \ell} \sqrt{1 - \varepsilon^2} + \sqrt{\ell} \varepsilon \right)^2 = \left(\sqrt{\ell} \sqrt{1 - \varepsilon^2} - \sqrt{1 - \ell} \varepsilon \right)^2$$

proves the desired bound in this case. If $\varepsilon \geq \sqrt{\ell}$, the expression inside the positive part is nonpositive, and the bound is zero. Rescaling x proves, in every case,

$$\|P_B x\|^2 \geq \left(\sqrt{\ell} \sqrt{1 - \varepsilon^2} - \sqrt{1 - \ell} \varepsilon \right)_+^2 \|x\|^2 \quad (x \in V). \quad (\text{IFE10})$$

If $K = \{0\}$, the estimate follows directly from $P_B = I$; zero components in the displayed decompositions follow by their limiting values. This also covers $\ell = 1$.

The bound is sharp over all data satisfying IFE8. In an auxiliary Euclidean two-plane set $K = \mathbb{C}e_1$, $B = \mathbb{C}e_2$, and $u = \sqrt{1 - \ell}e_1 + \sqrt{\ell}e_2$. If $0 \leq \varepsilon < \sqrt{\ell}$, take

$$x = \sqrt{1 - \varepsilon^2} u + \varepsilon(\sqrt{\ell}e_1 - \sqrt{1 - \ell}e_2), \quad U = \mathbb{C}u, \quad V = \mathbb{C}x. \quad (\text{IFE11})$$

The two terms are orthogonal, x is unit, the distance condition is exactly ε , and its observed coordinate equals the positive expression in IFE10. For $\varepsilon \geq \sqrt{\ell}$, choose $V = K$; its distance from U is $\sqrt{\ell} \leq \varepsilon$, and its observed fraction is zero. Thus the transition at $\varepsilon = \sqrt{\ell}$ is also optimal. These examples test the inequality; they do not replace the original arithmetic subspaces.

3.3 3. Exact polynomial division and the complete slow singular space

Let $E = \mathbb{C}[y]/(Q_k)$, $\deg Q_k = q$, with multiplication $M[f] = [yf]$, and keep the full source polynomial space $\mathcal{P}_N$ with its original positive metric. Let $J_N : \mathcal{P}_N \rightarrow E$ be the original value map, and let $\hat{R}_N : E \rightarrow \mathcal{P}_N$ be its minimum section. In the attained metric G_N , $\hat{R}_N$ is an isometry and J_N is contractive.

Take a monic polynomial p , $1 \leq d = \deg p \leq q$, coprime to Q_k , with $N \geq q - 1$. Polynomial division gives $f = p\mathfrak{Q}_p f + \mathfrak{R}_p f$, where $\deg \mathfrak{R}_p f < d$. If $N < d$, the quotient map is zero; among the retained endpoints this is exactly $N = q - 1, d = q$. Otherwise it maps to $\mathcal{P}_{N-d}$, with its native source metric.

Since $p(M)$ is invertible, applying the value map and its inverse to the division of $\hat{R}_N x$ proves

$$X := p(M)^{-1} = B_p + F_p, \quad B_p = J_N \mathfrak{Q}_p \hat{R}_N, \quad F_p = U_p T_p, \quad T_p = \mathfrak{R}_p \hat{R}_N, \quad U_p r = [r/p]. \quad (\text{IFE12})$$

The last map is injective: if $[r/p] = 0$, multiplication by the invertible class $[p]$ gives $[r] = 0$, and $\deg r < d \leq q$ forces $r = 0$. Thus $\dim \text{ran } U_p = d$, while $\text{rank } F_p \leq d$. With $D_p = \|\mathfrak{Q}_p\|$, contractivity and isometry give $\|B_p\| \leq D_p$. No rational function has been assigned a source integral here.

Let $\sigma_1(X) \geq \dots \geq \sigma_q(X) > 0$, and let V_d be a top d left singular space, with any admissible choice inside a tied singular eigenspace. Expanding a vector $x \in V_d$ in left singular vectors gives its unique preimage $u = X^{-1}x$, with

$$\|u\| \leq \|x\|/\sigma_d(X), \quad \|(I - P_{\text{ran } U_p})x\| = \|(I - P_{\text{ran } U_p})B_p u\| \leq D_p \|u\|. \quad (\text{IFE13})$$

Thus the distance hypothesis of IFE8 holds with $\varepsilon_p = \min(1, D_p/\sigma_d(X))$. On $\ker F_p$, a subspace of dimension at least $q - d$, the identity IFE12 gives $X = B_p$. The singular-value minimum-over-subspaces formula therefore yields

$$\sigma_{d+1}(X) \leq D_p \quad (d < q). \quad (\text{IFE14})$$

For an exact observation bound $\|P_B u\|^2 \geq \ell_p \|u\|^2$ on $\text{ran } U_p$, IFE10 proves the UFR16 factor

$$\ell_{\text{sharp}} = \left(\sqrt{\ell_p} \sqrt{1 - \varepsilon_p^2} - \sqrt{1 - \ell_p} \varepsilon_p \right)_+^2. \quad (\text{IFE15})$$

To verify the heat operator's singular orientation, write an original-metric singular decomposition $Xv_j = \sigma_j u_j$. Then $p(M)u_j = \sigma_j^{-1} v_j$, so $p(M)^\dagger p(M)u_j = \sigma_j^{-2} u_j$. The top left singular vectors of X , exactly the vectors used in IFE13, are therefore the slow heat eigenvectors. For any orthogonal projection P ,

$$\text{Tr}(P e^{-\tau p(M)^\dagger p(M)}) = \sum_{j=1}^q e^{-\tau/\sigma_j^2} \|Pu_j\|^2. \quad (\text{IFE16})$$

The weights for P_B on the first d vectors are at least ℓ_{sharp} , and those for P_K are at most $1 - \ell_{\text{sharp}}$. All remaining weights lie in $[0, 1]$. Together with IFE14, this proves both UFR17 bounds for every $\tau \geq 0$, with the tail $(q - d)e^{-\tau/D_p^2}$ when $d < q$. If $D_p = 0$, IFE12 and invertibility of X imply $q = \text{rank } F_p \leq d$, so $d = q$ and the tail is omitted. No division by zero is necessary.

3.4 4. The small matrix and the sufficient positive guard

In the original coefficient coordinates set

$$H_p = U_p^* G_N U_p > 0, \quad S_p = H_p^{1/2} T_p G_N^{-1} T_p^* H_p^{1/2} \succeq 0. \quad (\text{IFE17})$$

The map $U_p H_p^{-1/2} : \mathbb{C}^d \rightarrow (E, G_N)$ is an isometry. Hence the singular values of $F_p = U_p T_p$ are those of $H_p^{1/2} T_p : (E, G_N) \rightarrow \mathbb{C}^d$, with extra zeros in the q -dimensional presentation if needed. Its product with its metric adjoint is exactly S_p . Therefore the d eigenvalues of S_p are the squares of the first d singular values of F_p , including zeros. In particular

$$\sigma_d(F_p) = \sqrt{\mu_d}, \quad \mu_d = \lambda_{\min}(S_p) \geq 0. \quad (\text{IFE18})$$

The singular-value maximum-over-subspaces formula and $\|X - F_p\| \leq D_p$ give

$$\sigma_d(X) \geq \sigma_d(F_p) - D_p = \sqrt{\mu_d} - D_p. \quad (\text{IFE19})$$

Indeed, on a top d right singular space of F_p , the triangle inequality gives $\|Xz\| \geq \|F_p z\| - D_p \|z\| \geq (\sigma_d(F_p) - D_p) \|z\|$; maximizing over such subspaces gives IFE19. A negative right side is harmless.

If

$$\boxed{\sqrt{\mu_d} > D_p(1 + \ell_p^{-1/2})}, \quad (\text{IFE20})$$

then $\sqrt{\mu_d} - D_p > 0$, and

$$\varepsilon_p \leq \widehat{\varepsilon}_p := \frac{D_p}{\sqrt{\mu_d} - D_p} < \sqrt{\ell_p}. \quad (\text{IFE21})$$

The factor in IFE15 is decreasing as a function of ε before its zero, so replacing ε_p by this larger explicit bound preserves the lower estimate and makes it strictly positive. This proves UFR19 and the resulting heat lower bound. It also proves $\mu_d > 0$, including the case $D_p = 0$.

Positive definiteness before IFE20 cannot be inferred merely from the division identity. An auxiliary exact polynomial example demonstrates the missing inference. Take $Q(y) = y - 1$, $q = 1$, $N = 1$, and a probability source of mean zero and second

moment one. Its metric on $a + by$ is $|a|^2 + |b|^2$. The attained metric at the value $f(1)$ is $G_N = 1/2$, with minimum section $\hat{R}_N x = x(1 + y)/2$. For $p(y) = y + 1$, coprime to Q , division of every minimum lift has zero remainder. Thus $T_p = 0$ and $S_p = 0$, while $X = p(M)^{-1} = 1/2$ remains invertible. This is a test of the general finite reasoning, not a replacement of the programme polynomial. The correction is to retain zero eigenvalues until the actual guard is checked.

3.5 5. Exact rational recovery and its degree guard

Keep the physical coordinate S , the complete upper relation χ of degree q , the complete lower relation $\hat{\chi}$ of degree q' , and shifts satisfying $\hat{\chi}(S) \mid \chi(S + b_j)$. Set

$$\mathcal{T}_A f(S) = \sum_{j=1}^J a_j f(S + b_j), \quad E_A(z) = \sum_j a_j e^{b_j z}, \quad v = \text{ord}_0 E_A < \infty, \quad \mu_h = \sum_j a_j b_j^h. \quad (\text{IFE22})$$

Thus $\mu_h = 0$ for $h < v$, and $\mu_v \neq 0$. Let D be monic of degree $1 \leq d \leq q$, coprime to χ ; set $L = Jd$, $\mathcal{B} = \prod_j D(S + b_j)$, and $U_D p = [p/D]$ for $\deg p < d$.

For a nonzero such numerator, write $n_0 = d - \deg p \geq 1$ and $p/D = cS^{-n_0} + O(S^{-n_0-1})$, $c \neq 0$. Expanding each translated Laurent monomial at infinity shows that the first surviving coefficient of its sum is

$$\mathcal{T}_A(p/D) = (-1)^v \binom{n_0 + v - 1}{v} \mu_v c S^{-n_0-v} + O(S^{-n_0-v-1}). \quad (\text{IFE23})$$

All earlier coefficients vanish because their moments have index below v . Contributions from later Laurent monomials to the coefficient displayed in IFE23 also have lower moment index and vanish. Its coefficient is nonzero. This proves injectivity of $\mathcal{T}_A$ on nonzero proper rational functions.

The complete numerator $N_p = \mathcal{B} \mathcal{T}_A(p/D)$ is a polynomial. Since $\mathcal{B}$ is monic of degree L , IFE23 proves the exact degree formula

$$\deg N_p = L - n_0 - v \leq L - v - 1. \quad (\text{IFE24})$$

In particular the upper bound is attained for $p = S^{d-1}$. A negative value on the right for a nonzero numerator would contradict IFE23 and polynomiality, so no inconsistent negative-degree case is introduced.

A Bézout identity $DP + \chi R = 1$, shifted by b_j and reduced modulo $\hat{\chi}$, proves that every $D(S + b_j)$ is a unit of the complete lower quotient. Therefore $\mathcal{B}$ is a unit there, including all multiple-root jets. The same shifted identity proves the exact relation

$$M_{\mathcal{B}}^{(-)} C_A U_D p = [N_p]_{\hat{\chi}}. \quad (\text{IFE25})$$

This establishes the rational action from the original polynomial conductor, without evaluating only selected roots.

Let T_D be the L -by- d coefficient matrix of the numerators N_{S^ℓ} , $0 \leq \ell < d$. Injectivity in IFE23 gives full column rank, so $T_D^* T_D > 0$. If

$$Jd - v = L - v \leq q', \quad (\text{IFE26})$$

then every N_p has degree below q' by IFE24 and is already its unique lower remainder. Taking its first L coefficient rows, with zero padding or truncation of the lower-remainder coordinate space as required, returns exactly $T_D p$. Hence

$$W_D = (T_D^* T_D)^{-1} T_D^* \text{coef}_{<L} M_B^{(-)}, \quad \boxed{W_D C_A U_D = I_d}. \quad (\text{IFE27})$$

This proves UFR10 even if $q' < L$. The last v numerator rows vanish exactly. The condition IFE26 is sharp for all these numerators being unreduced remainders, because IFE24 attains degree $L - v - 1$. Beyond IFE26 a conductor may still be injective for other reasons, but that conclusion does not follow from this unreduced-numerator argument.

For completeness, retain the original observation Λ , with $\ker \Lambda \subseteq \ker C_A$, the positive metric G , and $Q = (\Lambda G^{-1} \Lambda^*)^{-1}$, $\Gamma_A = C_A G^{-1} C_A^*$, $H_D = U_D^* G U_D > 0$. The containment of kernels gives a unique map on the observation image with $C_A = R_A \Lambda$. Every lift of an observed value is therefore a lift of its conductor value. The minimum source norm on the latter fibre is $z^* \Gamma_A^+ z$ for $z \in \text{ran } C_A$, as follows by the minimum lift $G^{-1} C_A^* \Gamma_A^+ z$ and orthogonality to $\ker C_A$. Consequently

$$U_D^* \Lambda^* Q \Lambda U_D \succeq (C_A U_D)^* \Gamma_A^+ (C_A U_D). \quad (\text{IFE28})$$

IFE27 gives $\|C_A U_D a\| \geq \|a\| / \|W_D\|$. On its positive range, $\Gamma_A^+ \succeq \lambda_{\max}(\Gamma_A)^{-1} I$, while $H_D \preceq \lambda_{\max}(H_D) I$. Combining these inequalities proves the UFR12 factor

$$U_D^* \Lambda^* Q \Lambda U_D \succeq \ell_D H_D, \quad \ell_D = \frac{1}{\lambda_{\max}(\Gamma_A) \|W_D\|^2 \lambda_{\max}(H_D)} > 0. \quad (\text{IFE29})$$

Injectivity in IFE27 ensures a nonzero conductor image, so the denominator is positive. The observed minimum is at most the original norm, which forces $\ell_D \leq 1$. Thus the exact original-metric factor required in IFE8 is available on the stated conductor guard, with all maps and source terms retained.

The application to the monic y -filter in IFE12 has the following exact physical-coordinate map. Retain $S = s_0 + iy$, $s_0 = k/2$, and put

$$D(S) = i^d p((S - s_0)/i), \quad (Tr)(S) = i^d r((S - s_0)/i). \quad (\text{IFE30})$$

The polynomial D is monic of degree d , and T is invertible on the degree-below- d numerator space. Under the exact algebra identification between the original y and physical- S quotients, $\Phi([f(y)]) = [f((S - s_0)/i)]$,

$$\Phi(U_p r) = U_D (Tr), \quad \frac{i^d r((S - s_0)/i)}{i^d p((S - s_0)/i)} = \frac{r((S - s_0)/i)}{p((S - s_0)/i)}. \quad (\text{IFE31})$$

The complete relations correspond under the same substitution; their nonzero leading factors do not change their ideals. Transporting the original metric through this exact coordinate identification gives $H_p = T^* H_D T$, and the observed Gram transforms by the same congruence. Therefore IFE29 gives $\|P_B u\|^2 \geq \ell_D \|u\|^2$ on $\text{ran } U_p$, with the exact scalar $\ell_p = \ell_D$. No physical-coordinate scale or numerator coefficient has been dropped.

The independent conclusions are UFR6, the sharp UFR14, UFR15–17, the corrected semidefinite form of UFR18, the sufficient positive guard UFR19, and the full unreduced-numerator guard in UFR9–12. None requires replacing the original filter by a compressed observation action.

Part IV

Complete observed heat and memory proofs

4 Uniform observed heat with full memory and the original metric

Date: 2026-09-21. This independent derivation verifies equations (18)–(23) and (33)–(51) of the supplied **Uniform control of the original source families** intake. Its unchanged text is retained in `USER_INTAKE_VERBATIM.txt`. All results below concern the original finite metric spaces and maps. Orthonormal coordinates are used only through explicitly stated isometries. No kernel-invariance assumption is imposed.

4.1 1. Original metric, attained quotient, and exact block maps

Let E be a complex vector space of dimension q , with positive Hermitian metric G , and let $\Lambda : E \rightarrow B$ be onto, where $n = \dim B$. Let $K = \ker \Lambda$, $m = \dim K = q - n$, and define

$$Q = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q.$$

Then

$$\Lambda L = I_B, \quad L^* G L = Q.$$

For $k \in K$ and $b \in B$,

$$k^* G L b = k^* \Lambda^* Q b = (\Lambda k)^* Q b = 0.$$

Thus L is the exact isometry from (B, Q) onto $K^\perp_G$; $P_B = L \Lambda$ is the G -orthogonal projection onto that image and $P_K = I - P_B$. For every operator $T : E \rightarrow E$, cyclicity of finite-dimensional trace gives

$$\text{Tr}_B(\Lambda T L) = \text{Tr}_E(P_B T). \quad (\text{UH1})$$

This identity retains the observation target and its minimum section.

Choose any isometric inclusion $I_K : (K, G|_K) \rightarrow (E, G)$. In the orthogonal sum $K \oplus LB$, the fixed original action $M : E \rightarrow E$ is

$$M = \begin{pmatrix} M_K & D \\ C & M_B \end{pmatrix}, \quad M_K = I_K^\dagger M I_K, \quad D = I_K^\dagger M L, \quad C = \Lambda M I_K, \quad M_B = \Lambda M L. \quad (\text{UH2})$$

Every dagger in these blocks uses the displayed domain and target metrics. In particular $D : B \rightarrow K$, $C : K \rightarrow B$, and C need not vanish. The full positive operator is

$$H = M^\dagger M = \begin{pmatrix} A & B_H \\ B_H^\dagger & D_H \end{pmatrix}, \quad (\text{UH3})$$

where

$$A = M_K^\dagger M_K + C^\dagger C, \quad B_H = M_K^\dagger D + C^\dagger M_B, \quad D_H = D^\dagger D + M_B^\dagger M_B. \quad (\text{UH4})$$

These are precisely the intake's complete blocks. Set $J_B = M_B^\dagger M_B$.

All subsequent eigenvalue, norm, and matrix calculations may be made after isometries of $(K, G|_K)$ and (B, Q) to Euclidean spaces. Their values are unchanged by those isometries.

4.2 2. Positive memory without a spectral-gap hypothesis

4.2.1 2.1 The exact Schur remainder

If $k \in \ker A$, positivity of H applied to (uk, b) , with arbitrary $u \in \mathbb{C}$, gives

$$0 \leq 2\Re(\bar{u} \langle k, B_H b \rangle) + \langle b, D_H b \rangle.$$

An unbounded choice of the phase and modulus of u forces $\langle k, B_H b \rangle = 0$ for every b . Therefore

$$\ker A \subseteq \ker B_H^\dagger, \quad \text{ran } B_H \subseteq \text{ran } A. \quad (\text{UH5})$$

Let A^+ denote its positive-range inverse, zero on $\ker A$. Completing the square, for every k, b ,

$$\begin{aligned} \langle (k, b), H(k, b) \rangle &= \langle k + A^+ B_H b, A(k + A^+ B_H b) \rangle \\ &\quad + \langle b, (D_H - B_H^\dagger A^+ B_H) b \rangle. \end{aligned}$$

Taking $k = -A^+ B_H b$ proves

$$T = B_H^\dagger A^+ B_H \succeq 0, \quad S = D_H - T \succeq 0, \quad T \preceq D_H. \quad (\text{UH6})$$

Let P_λ be the spectral projection of A at the distinct positive eigenvalue λ , and put

$$C_\lambda = \lambda^{-1} B_H^\dagger P_\lambda B_H \succeq 0.$$

Equation (UH5) yields

$$T = \sum_{\lambda>0} C_\lambda, \quad \mathcal{M}(t) := B_H^\dagger e^{-tA} B_H = \sum_{\lambda>0} \lambda e^{-t\lambda} C_\lambda. \quad (\text{UH7})$$

In particular no unrecorded memory remains at the zero eigenvalue.

For z off the negative spectrum of A , the exact memory pencil is

$$\begin{aligned} \mathcal{F}(z) &= zI_B + D_H - B_H^\dagger (zI_K + A)^{-1} B_H \\ &= zI_B + S + \sum_{\lambda>0} \frac{z}{z + \lambda} C_\lambda. \end{aligned} \quad (\text{UH8})$$

Whenever the full resolvent and these expressions are defined, block elimination gives

$$\Lambda(zI_E + H)^{-1} L = \mathcal{F}(z)^{-1}. \quad (\text{UH9})$$

For $\Re z > 0$, all expressions are defined because $H \succeq 0$. For $\Im z > 0$,

$$\Im \mathcal{F}(z) = (\Im z) \left[I_B + \sum_{\lambda>0} \frac{\lambda}{|z + \lambda|^2} C_\lambda \right] \succ 0. \quad (\text{UH10})$$

This also proves invertibility there. The matrices C_λ are never required to commute.

For completeness, define $Y(t) = \Lambda e^{-tH} L$ and let $X(t)$ be the kernel block of $e^{-tH} L$. The original block evolution is

$$X' = -AX - B_H Y, \quad X(0) = 0, \quad Y' = -B_H^\dagger X - D_H Y, \quad Y(0) = I_B.$$

Consequently

$$X(t) = - \int_0^t e^{-(t-u)A} B_H Y(u) du,$$

and the full observed equation is exactly

$$Y'(t) = -D_H Y(t) + \int_0^t \mathcal{M}(t-u) Y(u) du, \quad Y(0) = I_B. \quad (\text{UH11})$$

Both $D^\dagger D$ inside D_H and the full memory convolution remain.

4.2.2 2.2 Derivatives, integrals, and the half-plane bounds

For every integer $j \geq 0$,

$$(-1)^j \mathcal{M}^{(j)}(t) = \sum_{\lambda > 0} \lambda^{j+1} e^{-t\lambda} C_\lambda \succeq 0.$$

By the substitution $u = \lambda t$ and the elementary gamma integral,

$$\int_0^\infty t^j (-1)^j \mathcal{M}^{(j)}(t) dt = j! \sum_{\lambda > 0} C_\lambda = j! T \preceq j! D_H. \quad (\text{UH12})$$

Also $\sup_{u \geq 0} u^{j+1} e^{-u} = ((j+1)/e)^{j+1}$, so

$$t^{j+1} (-1)^j \mathcal{M}^{(j)}(t) \preceq ((j+1)/e)^{j+1} T. \quad (\text{UH13})$$

These estimates have no lower positive bound for λ .

Differentiating the finite rational expression (UH8), for $j \geq 1$,

$$(-1)^{j-1} (\mathcal{F}^{(j)}(z) - \mathbf{1}_{j=1} I_B) = j! \sum_{\lambda > 0} \frac{\lambda}{(z + \lambda)^{j+1}} C_\lambda \quad (z > 0).$$

Writing $u = \lambda/z$, the scalar maximum

$$\sup_{u \geq 0} \frac{u}{(1+u)^{j+1}} = \frac{j^j}{(j+1)^{j+1}}$$

occurs at $u = 1/j$. Hence

$$0 \preceq z^j (-1)^{j-1} (\mathcal{F}^{(j)}(z) - \mathbf{1}_{j=1} I_B) \preceq j! \frac{j^j}{(j+1)^{j+1}} T. \quad (\text{UH14})$$

This proves every constant in intake equations (35)-(41).

4.2.3 2.3 Exact minimal memory and its original-space isometry

The original coupled kernel space is

$$K_{\text{coupled}} = \bigoplus_{\lambda > 0} \text{ran}(P_\lambda B_H).$$

Each summand lies in a distinct eigenspace of A . Its orthogonal complement in K is A -invariant and annihilated by $B_H^\dagger$. Thus the full H splits as its restriction to $K_{\text{coupled}} \oplus LB$, plus the original decoupled kernel block.

Let $R_\lambda = \text{ran } C_\lambda$, with its inherited metric, and define

$$J_\lambda = \lambda^{-1/2} P_\lambda B_H C_\lambda^{+1/2} : R_\lambda \rightarrow K.$$

Because $B_H^\dagger P_\lambda B_H = \lambda C_\lambda$,

$$J_\lambda^\dagger J_\lambda = C_\lambda^{+1/2} C_\lambda C_\lambda^{+1/2} = I_{R_\lambda}.$$

The image is exactly $\text{ran}(P_\lambda B_H)$, because multiplying by the positive-range inverse does not change that range. The direct sum $J = \bigoplus J_\lambda$ is therefore an isometry onto K_{coupled} . Let

$$A_{\min} = \bigoplus_{\lambda > 0} \lambda I_{R_\lambda}, \quad B_{\min} = \text{stack}_{\lambda > 0} \sqrt{\lambda} C_\lambda^{1/2}.$$

Then

$$AJ = JA_{\min}, \quad J_\lambda \sqrt{\lambda} C_\lambda^{1/2} = P_\lambda B_H C_\lambda^{+1/2} C_\lambda^{1/2} = P_\lambda B_H.$$

The last equality holds since C_λ and $P_\lambda B_H$ have the same kernel. Summing proves $JB_{\min} = B_H$. Therefore

$$\begin{pmatrix} A_{\min} & B_{\min} \\ B_{\min}^\dagger & D_H \end{pmatrix}$$

is the exact isometric restriction of H to its coupled block, and is positive. It gives the identical $Y(t)$, memory, and compressed resolvent.

Its kernel-state dimension is

$$r_{\text{mem}} = \sum_{\lambda > 0} \text{rank } C_\lambda.$$

To prove minimality, take any finite selfadjoint realization $\tilde{B}^\dagger(zI + \tilde{A})^{-1}\tilde{B}$ of the same memory transform. At the pole $-\lambda$, its residue is $\tilde{B}^\dagger \tilde{P}_\lambda \tilde{B}$; equality of the rational transforms forces this residue to equal λC_λ . Therefore

$$\dim \text{ran } \tilde{P}_\lambda \geq \text{rank } C_\lambda.$$

The eigenspaces for distinct poles are disjoint. Summing proves the required lower bound r_{mem} . Extra states with zero residue are decoupled from the memory and cannot lower it.

4.3 3. Uniform heat comparison by the original coupling and leakage ranks

Put

$$H_0 = \text{diag}(A, D_H), \quad O = H - H_0 = \begin{pmatrix} 0 & B_H \\ B_H^\dagger & 0 \end{pmatrix}, \quad r_c = \text{rank } B_H.$$

Let

$$Q_{\text{mix}}(t) = \text{Tr } e^{-tH} - \text{Tr } e^{-tA} - \text{Tr } e^{-tD_H}.$$

4.3.1 3.1 Nonnegative trace gaps on each original block

If v is a unit eigenvector of A , regard it as $(v, 0) \in E$. Its spectral measure μ_v for H is a probability measure. The scalar convexity of $x \mapsto e^{-tx}$ gives

$$\langle v, e^{-tH} v \rangle = \int e^{-tx} d\mu_v(x) \geq e^{-t \int x d\mu_v(x)} = e^{-t \langle v, H v \rangle}.$$

Sum over an orthonormal eigenbasis of A . The result is

$$\text{Tr}(P_K e^{-tH}) - \text{Tr } e^{-tA} \geq 0.$$

An eigenbasis of D_H similarly gives

$$a(t) := \text{Tr}(P_B e^{-tH}) - \text{Tr } e^{-tD_H} \geq 0.$$

Thus

$$0 \leq a(t) \leq Q_{\text{mix}}(t). \quad (\text{UH15})$$

This is a trace argument; an operator order between the two observed heat matrices is not used.

4.3.2 3.2 Signed rank bound, with its full interlacing proof

The nonzero eigenvalues of O are the pairs $\pm\sigma_j(B_H)$, $1 \leq j \leq r_c$. Hence its positive and negative parts O_+ and O_- each have rank r_c .

Here is the needed rank fact. If $X, Y \succeq 0$, $X = Y - R$, $R \succeq 0$, and $\text{rank } R \leq r$, order their q eigenvalues increasingly. Min-max gives

$$\lambda_j(X) \leq \lambda_j(Y), \quad \lambda_j(X) \geq \lambda_{j-r}(Y) \quad (j > r).$$

For a decreasing function $0 \leq f \leq 1$ on $[0, \infty)$,

$$\sum_j f(\lambda_j(X)) \leq r + \sum_{j=1}^{q-r} f(\lambda_j(Y)) \leq r + \sum_{j=1}^q f(\lambda_j(Y)). \quad (\text{UH16})$$

The first eigenvalue inequality also gives nonnegative trace difference.

Apply (UH16) to $X = H$, $Y = H_0 + O_+$, and $R = O_-$. Both endpoint matrices are positive, and $Y = H + O_-$. Then

$$\text{Tr } e^{-tH} \leq r_c + \text{Tr } e^{-t(H_0+O_+)} \leq r_c + \text{Tr } e^{-tH_0},$$

where the second inequality follows from min-max and $O_+ \succeq 0$. Therefore

$$0 \leq Q_{\text{mix}}(t) \leq r_c. \quad (\text{UH17})$$

The signed ranks give r_c , not the total rank $2r_c$.

4.3.3 3.3 The exact small-time quadratic estimate

For $0 \leq s \leq 1$, $H_s = H_0 + sO = (1-s)H_0 + sH \succeq 0$. Put $\phi(s) = \text{Tr } e^{-tH_s}$. Differentiating the exponential by its convergent power series and cyclically moving factors in the trace gives

$$\phi'(s) = -t \text{Tr}(O e^{-tH_s}),$$

and a second differentiation gives

$$\phi''(s) = t^2 \int_0^1 \text{Tr}(O e^{-(1-u)tH_s} O e^{-utH_s}) du. \quad (\text{UH18})$$

In an orthonormal eigenbasis of H_s , the integrand is

$$\sum_{i,j} |O_{ij}|^2 e^{-t[(1-u)\lambda_j + u\lambda_i]},$$

which lies between 0 and $\|O\|_{\text{HS}}^2$. Since $\phi'(0) = 0$ and $\|O\|_{\text{HS}}^2 = 2\|B_H\|_{\text{HS}}^2$,

$$Q_{\text{mix}}(t) = \int_0^1 (1-s)\phi''(s) ds \leq t^2 \|B_H\|_{\text{HS}}^2.$$

Together with (UH17),

$$0 \leq Q_{\text{mix}}(t) \leq \min\{r_c, t^2 \|B_H\|_{\text{HS}}^2\}. \quad (\text{UH19})$$

4.3.4 3.4 Leakage and the complete observed error

Since $D_H = J_B + D^\dagger D$, let

$$b(t) = \text{Tr} e^{-tJ_B} - \text{Tr} e^{-tD_H} \geq 0.$$

The rank fact (UH16), with $r_D = \text{rank } D$, gives $b(t) \leq r_D$. Also

$$b(t) = t \int_0^1 \text{Tr}(D^\dagger D e^{-t(J_B + sD^\dagger D)}) ds.$$

The exponential is a positive contraction, so

$$0 \leq b(t) \leq \min\{r_D, t\|D\|_{\text{HS}}^2\}. \quad (\text{UH20})$$

The actual observed error is exactly $a(t) - b(t)$. Equations (UH15), (UH19), and (UH20) prove

$$\boxed{-\min\{r_D, t\|D\|_{\text{HS}}^2\} \leq \text{Tr}_B(\Lambda e^{-tM^\dagger M} L) - \text{Tr}_B e^{-tM_B^\dagger M_B} \leq \min\{r_c, t^2\|B_H\|_{\text{HS}}^2\}.} \quad (\text{UH21})$$

Because $D : B \rightarrow K$ and $B_H : B \rightarrow K$, both ranks are at most $\min(m, n)$. In particular,

$$\boxed{\sup_{t \geq 0} \left| \text{Tr}_B(\Lambda e^{-tM^\dagger M} L) - \text{Tr}_B e^{-tM_B^\dagger M_B} \right| \leq \max(r_D, r_c) \leq \min(m, n).} \quad (\text{UH22})$$

This slightly strengthens the intake's dimension bound when the observed target is smaller than the kernel. No step sets $C = 0$, drops $D^\dagger D$, or discards memory.

For a source family where the established original kernel rank is $m_k = 8k - 16$, (UH22) yields the claimed $8k - 16$ bound at every allowed cutoff, metric, and heat time. If additionally $q_k = (k + 1)^2$, then

$$0 \leq \frac{\sup_{t \geq 0} |\text{observed error}|}{q_k} \leq \frac{8k - 16}{(k + 1)^2} \rightarrow 0.$$

These last two arithmetic dimension identities are inputs from the original family; the finite-dimensional theorem does not derive them independently.

4.3.5 3.5 Sharpness and a finite operator-order counterexample

Both signs in (UH21) are needed. In the Euclidean space $\mathbb{C}^2 = K \oplus B$, take

$$M_- = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $J_B = 0$, $H = \text{diag}(0, 1)$, and the observed error is

$$e^{-t} - 1 \rightarrow -1.$$

Thus the coefficient one in the dimension bound cannot be lowered.

For the positive sign, take

$$M_+ = \begin{pmatrix} 0 & 0 \\ \sqrt{a} & \sqrt{d} \end{pmatrix}, \quad a, d > 0.$$

Here $D = 0$, $C = \sqrt{a} \neq 0$, $J_B = d$, and

$$H = \begin{pmatrix} a & \sqrt{ad} \\ \sqrt{ad} & d \end{pmatrix}.$$

Since $H^2 = (a + d)H$,

$$\text{Tr}(P_B e^{-tH}) = \frac{a}{a+d} + \frac{d}{a+d} e^{-t(a+d)}.$$

The error tends to $a/(a+d)$, arbitrarily close to one when a/d grows. This example has a noninvariant kernel. Direct sums produce the corresponding sharp rank-scale examples.

The observed trace inequality (UH15) cannot be upgraded to operator order. At $t = \log 2$, take

$$M = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \\ -2 & -1 & 0 \end{pmatrix}, \quad H = M^* M = \begin{pmatrix} 5 & 2 & -2 \\ 2 & 1 & 0 \\ -2 & 0 & 4 \end{pmatrix}, \quad K = \mathbb{C}e_1.$$

The eigenvalues of H are 0, 3, 7, and $D_H = \text{diag}(1, 4)$. The spectral-projector formula

$$2^{-H} = \sum_{\lambda \in \{0, 3, 7\}} 2^{-\lambda} \prod_{\mu \neq \lambda} \frac{H - \mu I}{\lambda - \mu}$$

gives the exact rational difference

$$(2^{-H})_{BB} - 2^{-D_H} = \begin{pmatrix} 1523/5376 & -403/2688 \\ -403/2688 & 95/1344 \end{pmatrix}. \quad (\text{UH23})$$

Its determinant is $-211/86016 < 0$, while its trace is $1903/5376 > 0$. Thus it has one negative and one positive eigenvalue. This exact fixture explains why the scalar spectral-measure proof is essential.

4.4 4. Determinant mixing and the original observed pencil

For $z > 0$, define

$$g(z) = \log \det(zI_K + A) + \log \det(zI_B + D_H) - \log \det(zI_E + H).$$

The exact block determinant identity gives

$$g(z) = \log \det(zI_B + D_H) - \log \det \mathcal{F}(z). \quad (\text{UH24})$$

For positive numbers x, y ,

$$\log x - \log y = \int_0^\infty (e^{-ty} - e^{-tx}) dt/t.$$

One can prove this by differentiating in x , integrating e^{-tx} , and fixing the zero value at $x = y$. Apply it to the finite lists of positive eigenvalues after adding z . Equality of their total dimensions cancels the constant term at $t = 0$, and (UH19) gives absolute convergence. Hence

$$g(z) = \int_0^\infty e^{-zt} \frac{Q_{\text{mix}}(t)}{t} dt \geq 0. \quad (\text{UH25})$$

For $j \geq 1$, differentiation is justified by the same bounds near zero and exponential decay at infinity, giving

$$(-1)^j g^{(j)}(z) = \int_0^\infty e^{-zt} t^{j-1} Q_{\text{mix}}(t) dt \geq 0.$$

Consequently

$$0 \leq g(z) \leq \|B_H\|_{\text{HS}}^2 / z^2, \quad (\text{UH26})$$

and

$$0 \leq (-1)^j g^{(j)}(z) \leq \min \left\{ \frac{r_c(j-1)!}{z^j}, \frac{\|B_H\|_{\text{HS}}^2(j+1)!}{z^{j+2}} \right\}. \quad (\text{UH27})$$

For the logarithmic rank bound, put $R_z = B_H^\dagger(zI_K + A)^{-1}B_H$. It is positive of rank at most r_c , and

$$zI_B + D_H = \mathcal{F}(z) + R_z, \quad \mathcal{F}(z) \succeq zI_B$$

by (UH8). Thus

$$g(z) = \log \det(I_B + \mathcal{F}(z)^{-1/2} R_z \mathcal{F}(z)^{-1/2}).$$

There are at most r_c nonzero eigenvalues in its added matrix. Moreover

$$zI_B + D_H \preceq (1 + \|D_H\|/z) \mathcal{F}(z),$$

since the right side is at least $(z + \|D_H\|)I_B$. Therefore each corresponding multiplicative eigenvalue is at most $1 + \|D_H\|/z$, and

$$g(z) \leq r_c \log(1 + \|D_H\|/z).$$

Applying the same block determinant identity in the other order proves

$$0 \leq g(z) \leq r_c \log \left(1 + \frac{\min(\|A\|, \|D_H\|)}{z} \right). \quad (\text{UH28})$$

The original leakage determinant is

$$\ell(z) = \log \det(zI_B + D_H) - \log \det(zI_B + J_B).$$

It has the exact form

$$\ell(z) = \log \det \left(I_B + (zI_B + J_B)^{-1/2} D^\dagger D (zI_B + J_B)^{-1/2} \right) \geq 0.$$

At most r_D eigenvalues are nonzero, and each is at most $\|D\|^2/z$. Thus

$$0 \leq \ell(z) \leq r_D \log(1 + \|D\|^2/z). \quad (\text{UH29})$$

Subtracting (UH24) gives the original comparison

$$\log \det \mathcal{F}(z) - \log \det(zI_B + J_B) = \ell(z) - g(z). \quad (\text{UH30})$$

The two nonnegative terms are retained. The usable asymmetric bound is

$$\boxed{-r_c \log \left(1 + \frac{\min(\|A\|, \|D_H\|)}{z} \right) \leq \ell(z) - g(z) \leq r_D \log(1 + \|D\|^2/z).} \quad (\text{UH31})$$

Every block compression has norm bounded by the full positive operator norm, and D is a compression of M . Therefore

$$\|A\|, \|D_H\|, \|D\|^2 \leq \|M\|_G^2,$$

and

$$|\ell(z) - g(z)| \leq \max(r_D, r_c) \log(1 + \|M\|_G^2/z) \leq m \log(1 + \|M\|_G^2/z). \quad (\text{UH32})$$

This proves intake equation (50), with a sharper version in (UH31).

The intake additionally supplies, for its joint-source metric, $\log^+ \|M\|_{G_N^J} = O_h(k \log^2(q+2))$. For fixed $z > 0$,

$$\log(1 + \|M\|^2/z) \leq \log(1 + z^{-1}) + 2 \log^+ \|M\|.$$

Together with $m = O(k)$, this gives

$$|\ell(z) - g(z)| = O_h(k^2 \log^2(q+2)).$$

If $q = (k+1)^2$, division by kq gives

$$O_h\left(\frac{k \log^2((k+1)^2 + 2)}{(k+1)^2}\right) \rightarrow 0.$$

The asymptotic implication is proved here. The source-family norm estimate itself remains the cited original arithmetic input, and is not proved or altered by the finite-dimensional argument.

4.5 5. Metric changes and complete heat derivatives

4.5.1 5.1 Endpoint comparison in the original fixed coefficient space

Suppose two positive metrics G_x, G_y on the same coefficient space obey

$$e^{-b}G_x \preceq G_y \preceq e^{-a}G_x, \quad a \leq b. \quad (\text{UH33})$$

For every nonzero coefficient vector v ,

$$e^{-(b-a)} \frac{\|Mv\|_{G_x}^2}{\|v\|_{G_x}^2} \leq \frac{\|Mv\|_{G_y}^2}{\|v\|_{G_y}^2} \leq e^{b-a} \frac{\|Mv\|_{G_x}^2}{\|v\|_{G_x}^2}.$$

The min-max formula for the eigenvalues of the positive squared action, applied to the same coefficient subspaces at both endpoints, therefore gives

$$e^{-(b-a)/2} s_j(M; G_x) \leq s_j(M; G_y) \leq e^{(b-a)/2} s_j(M; G_x). \quad (\text{UH34})$$

This compares the exact metric singular values, with the identity of coefficient spaces as the linking map.

Let $r = \text{rank } M$, $\delta = b - a$. The $q - r$ zero singular directions have the same heat contribution at both endpoints. For each positive squared singular value pair λ, μ , $|\log \mu - \log \lambda| \leq \delta$. The function $s \mapsto e^{-te^s}$ has absolute derivative $te^s e^{-te^s} \leq 1/e$. Summing yields

$$\sup_{t \geq 0} |\text{Tr } e^{-tH_y} - \text{Tr } e^{-tH_x}| \leq r\delta/e. \quad (\text{UH35})$$

For $z > 0$, the derivative of $\log(z + e^s)$ lies in $(0, 1)$, giving

$$|\log \det(zI + H_y) - \log \det(zI + H_x)| \leq r\delta. \quad (\text{UH36})$$

These are intake (18)–(19).

A sharper bounded finite-change version of (UH35) is also available. Put

$$\Psi(0) = 0, \quad \Psi(\delta) = (1 - e^{-\delta}) \exp \left[-\frac{\delta}{e^\delta - 1} \right] \quad (\delta > 0).$$

For $\alpha = e^\delta > 1$, the maximum of $e^{-u} - e^{-\alpha u}$, $u \geq 0$, occurs at $u = (\log \alpha)/(\alpha - 1)$, and equals $\Psi(\delta)$. A smaller allowed ratio gives a smaller maximum because $e^{-u} - e^{-\alpha u}$ increases pointwise in α . Hence

$$\sup_{t \geq 0} |\text{Tr } e^{-tH_y} - \text{Tr } e^{-tH_x}| \leq r\Psi(\delta) \leq r \min(\delta/e, 1). \quad (\text{UH37})$$

No norm bound for M is needed.

4.5.2 5.2 The compatible frame for logarithmic source paths

Use the original positive covariance family

$$C(x) = G_o^{-1} + \sum_i e^{x_i} \Omega_i, \quad G(x) = C(x)^{-1}, \quad \Omega_i \succeq 0.$$

Along $x(s) = x_0 + sv$, put

$$v_- = \min(0, v_1, \dots, v_p), \quad v_+ = \max(0, v_1, \dots, v_p), \quad \Delta_v = v_+ - v_-.$$

Since

$$C' = \sum_i v_i e^{x_i} \Omega_i,$$

the positive baseline and source terms imply

$$v_- C \preceq C' \preceq v_+ C.$$

Differentiating $GC = I$, $G' = -GC'G$, gives

$$-v_+ G \preceq G' \preceq -v_- G. \quad (\text{UH38})$$

Choose $F(s_0)^* G(s_0) F(s_0) = I$ and solve the linear matrix equation

$$F' = -\frac{1}{2} G^{-1} G' F. \quad (\text{UH39})$$

Its fundamental solution is invertible. Direct differentiation gives

$$(F^* G F)' = -\frac{1}{2} F^* G' F + F^* G' F - \frac{1}{2} F^* G' F = 0.$$

Thus $F^* G F = I$, $F^{-1} = F^* G$, and

$$B(s) = F^* G' F = B^*, \quad -v_+ I \preceq B \preceq -v_- I. \quad (\text{UH40})$$

The spectral width of B is at most Δ_v .

Write the fixed original action in this frame as $\widetilde{M} = F^{-1} M F$. Then $F^{-1} F' = -B/2$, so

$$\widetilde{M}' = \frac{1}{2} [B, \widetilde{M}]. \quad (\text{UH41})$$

All following matrices are in this frame; the original coefficient action M is unchanged. With $\widetilde{H} = \widetilde{M}^* \widetilde{M}$,

$$\widetilde{H}' = \widetilde{M}^* B \widetilde{M} - \frac{1}{2} (B \widetilde{H} + \widetilde{H} B). \quad (\text{UH42})$$

Let c be the midpoint of the extreme eigenvalues of B , and $B_0 = B - cI$. The scalar terms cancel identically in (UH42), and

$$\|B_0\| \leq \Delta_v/2. \quad (\text{UH43})$$

This subtraction acts on the connection matrix only; it does not replace the original arithmetic action.

4.5.3 5.3 Both divided-difference constants

For $x, y \geq 0$, let

$$d_t(x, y) = \begin{cases} (e^{-tx} - e^{-ty})/(x - y), & x \neq y, \\ -te^{-tx}, & x = y. \end{cases}$$

Then

$$\sqrt{xy} |d_t(x, y)| \leq 1/e, \quad \frac{x+y}{2} |d_t(x, y)| \leq 1/2. \quad (\text{UH44})$$

For the first inequality, if $x, y > 0$, the function $u \mapsto e^{-te^u}$ is $1/e$ -Lipschitz, so

$$|e^{-tx} - e^{-ty}| \leq |\log x - \log y|/e.$$

Writing $x = \rho e^a, y = \rho e^{-a}$ proves

$$\frac{\sqrt{xy} |\log x - \log y|}{|x - y|} = \frac{|a|}{|\sinh a|} \leq 1.$$

The diagonal limit is $txe^{-tx} \leq 1/e$; if either x or y is zero, the product is zero.

For the second inequality take $tx = b + d, ty = b, b, d \geq 0$. For $d > 0$, twice the required left side is

$$e^{-b}(1 + 2b/d)(1 - e^{-d}).$$

For $d \geq 2$, this decreases with b , so its maximum is $1 - e^{-d} < 1$. For $0 < d < 2$, its maximum occurs at $b = 1 - d/2$ and equals

$$\frac{2 \sinh(d/2)}{e^{d/2}} \leq \frac{2}{e} \sinh 1 = 1 - e^{-2} < 1.$$

Here $\sinh u/u$ increases for $u > 0$, because $u \cosh u - \sinh u$ has derivative $u \sinh u > 0$ and value zero at zero. The limits $d = 0$ and $t = 0$ follow by continuity. Both individual constants are sharp: the first at $tx = ty = 1$, the second as $ty = 0$ and $tx \rightarrow \infty$.

The derivative formula for a matrix exponential can be read directly from

$$\frac{d}{ds} e^{-t\tilde{H}} = - \int_0^t e^{-(t-u)\tilde{H}} \tilde{H}' e^{-u\tilde{H}} du.$$

In an eigenbasis of $\tilde{H}$, its ij -entry equals $d_t(\lambda_i, \lambda_j) \tilde{H}'_{ij}$.

Let $\tilde{M} = U\tilde{H}^{1/2}$ be its actual polar decomposition. Equation (UH42) gives

$$\tilde{H}'_{ij} = \sqrt{\lambda_i \lambda_j} (U^* B_0 U)_{ij} - \frac{1}{2} (\lambda_i + \lambda_j) (B_0)_{ij}.$$

Applying (UH44) entry by entry and the Hilbert-Schmidt triangle inequality proves

$$\left\| \frac{d}{ds} e^{-t\tilde{H}} \right\|_{\text{HS}} \leq e^{-1} \|U^* B_0 U\|_{\text{HS}} + \frac{1}{2} \|B_0\|_{\text{HS}} \leq \left(\frac{1}{e} + \frac{1}{2} \right) \frac{\sqrt{q}}{2} \Delta_v. \quad (\text{UH45})$$

This proves intake (20), including zero and repeated eigenvalues.

There is a useful rank refinement. Put $r = \text{rank } M$ and let P_+ be the support projection of $\tilde{H}$. The first term satisfies $\|U^* B_0 U\|_{\text{HS}} \leq \sqrt{r} \|B_0\|$. The second term has zero entries on $\ker \tilde{H} \times \ker \tilde{H}$. If B_{++} and B_{+0} are its blocks, the square of the norm of the remaining entries is

$$\|B_{++}\|_{\text{HS}}^2 + 2\|B_{+0}\|_{\text{HS}}^2 \leq 2r\|B_0\|^2,$$

because $\|B_{++}\|_{\text{HS}}^2 + \|B_{+0}\|_{\text{HS}}^2 = \|P_+ B_0\|_{\text{HS}}^2 \leq r\|B_0\|^2$. It is also at most $q\|B_0\|^2$. Thus

$$\left\| \frac{d}{ds} e^{-t\tilde{H}} \right\|_{\text{HS}} \leq \frac{\Delta_v}{2} \left(\frac{\sqrt{r}}{e} + \frac{\sqrt{\min(q, 2r)}}{2} \right). \quad (\text{UH46})$$

This refinement is zero when the original action is zero.

4.5.4 5.4 Moving fixed-kernel projections, with the exact sign

Let $K \subset E$ be fixed in original coefficient coordinates and let $P(s)$ be the Euclidean orthogonal projection onto $F(s)^{-1}K$. A fixed vector $k \in K$ has frame coordinates $u(s) = F(s)^{-1}k$, satisfying $u' = Bu/2$. Since $Pu = u$, differentiation gives

$$P'u = (I - P)u' = \frac{1}{2}(I - P)Bu.$$

Differentiating $P^2 = P$ gives zero diagonal blocks for P' ; differentiating $P^* = P$ makes its off-diagonal blocks adjoints. Hence

$$P' = \frac{1}{2}[(I - P)BP + PB(I - P)]. \quad (\text{UH47})$$

This is intake (22) with its positive sign. The complementary projection has the negative derivative.

Let $\ell = \text{rank } P$, $d = \min(\ell, q - \ell)$, and $T_P = (I - P)B_0P$. The nonzero eigenvalues of P' are $\pm\sigma_j(T_P)/2$. Therefore

$$\text{Tr}(P')_+ = \text{Tr}(P')_- = \frac{1}{2}\|T_P\|_1 \leq d\Delta_v/4. \quad (\text{UH48})$$

For any positive contraction Z ,

$$-\text{Tr}(P')_- \leq \text{Tr}(P'Z) \leq \text{Tr}(P')_+.$$

Apply this to $Z = e^{-t\tilde{H}}$. The same bound holds for the complementary projection.

Finally, for either the fixed-kernel projection or its complement,

$$\frac{d}{ds} \text{Tr}(Pe^{-t\tilde{H}}) = \text{Tr}(P'e^{-t\tilde{H}}) + \text{Tr}\left(P \frac{d}{ds} e^{-t\tilde{H}}\right).$$

Hilbert-Schmidt Cauchy-Schwarz gives the second term's absolute value at most $\sqrt{\ell}$ times (UH45). Therefore

$$\left| \frac{d}{ds} \text{Tr}(Pe^{-t\tilde{H}}) \right| \leq \Delta_v \left[\frac{\min(\ell, q - \ell)}{4} + \frac{1/e + 1/2}{2} \sqrt{\ell q} \right]. \quad (\text{UH49})$$

This is exactly intake (23). A stronger version replaces its second term by

$$\frac{\sqrt{\ell}}{2} \left(\frac{\sqrt{r}}{e} + \frac{\sqrt{\min(q, 2r)}}{2} \right). \quad (\text{UH50})$$

For the original fixed observation, (UH1) identifies this trace with $\text{Tr}_B(\Lambda e^{-tM^\dagger G M} L)$. Thus the bound includes the moving minimum section and both moving metric factors.

These projection formulas require the original subspace K to be fixed along the source-weight path; they do not require $MK \subseteq K$. If the observation itself varies, its additional subspace velocity must be added to (UH47). An unconstrained velocity has no dimension-only bound.

4.6 6. Status of the intake equations

The finite-dimensional claims (18)-(23) and (33)-(50) are valid with the complete original metric interpretation. Equation (51) follows from the supplied joint-source action-norm estimate and the original dimension growth; this derivation verifies that implication, not the separate arithmetic norm estimate.

The additional established refinements are (UH22), (UH31), (UH37), and (UH46)/(UH50). Equation (UH23) is an exact finite counterexample to an operator-order strengthening that the intake correctly does not claim. The fixture script and its output distinguish exact rational checks from finite-precision sampling. Neither substitutes for the complete proofs above.

Part V

Complete supplied uniform-family continuation

5 Uniform control of the original source families

The continuation now gives **uniform control of the complete finite metric family**, not just its determinant. It covers all fixed restrictions and attained quotients, all mixed derivatives in the source weights, the full observed heat operator with its memory term, and rational conductor observations with repeated or colliding poles.

One consequence directly concerns the original simple-quartet observation. Retain its original kernel $K_k = \ker \Lambda_k$, original metric G_N , original minimum section L_N , and

$$M = \frac{M_S - kI/2}{i}, \quad M_{B,N} = \Lambda_k M L_N.$$

Then, at every original cutoff and every heat time,

$$\sup_{\tau \geq 0} \left| \text{Tr} \left(\Lambda_k e^{-\tau M^\dagger G_N M} L_N \right) - \text{Tr} e^{-\tau M_{B,N}^\dagger M_{B,N}} \right| \leq 8k - 16. \quad (1)$$

Consequently,

$$\sup_{\tau \geq 0} \frac{1}{(k+1)^2} \left| \text{Tr} \left(\Lambda_k e^{-\tau M^\dagger G_N M} L_N \right) - \text{Tr} e^{-\tau M_{B,N}^\dagger M_{B,N}} \right| \rightarrow 0. \quad (2)$$

The complete memory and leakage terms are retained in the proof. No lower bound for a spectral gap, no estimate for the largest native singular value, and no assumption that K_k is invariant under M enters (1).

A second consequence is an exact complexity restriction on the original observation kernel. Let the actual conductor have J labelled shifts, lower-packet degree q' , and first nonzero symbol moment μ_v . For every nonzero $x \in K_k$, define its proper rational complexity by

$$d_{\text{rat}}(x) = \min \{ \deg D : D \text{ monic}, (D, \chi_k) = 1, x = [p/D], \deg p < \deg D \}.$$

Then

$$d_{\text{rat}}(x) \geq \left\lfloor \frac{q' + v}{J} \right\rfloor + 1. \quad (3)$$

This holds for arbitrary poles and all their multiplicities—not only the inverse powers at zero considered in the preceding calculation.

The repository inputs used here are the complete inverse-power conductor and observed-heat/memory proofs already retrieved at `fe22dc1dd51c633e82f93b731cbb6ec96b11ed64`. I have not verified a later revision in this continuation. The new work below extends those fixed sources; it does not claim that subsequent GitHub changes have been incorporated.

5.1 1. A common complex neighbourhood for every fixed subquotient

Start with the original finite arithmetic value space E , a positive metric $G_\circ$, and the actual positive covariance contributions of complete normal-source groups:

$$\Omega_1, \dots, \Omega_p \succeq 0.$$

Their construction retains the original cross Gram and positive-range inverse. In particular, the original single addition remains

$$\Omega = F_Z H_Z^+ F_Z^*,$$

with

$$H_Z = \Gamma - C_X^* H_X^{-1} C_X, \quad F_Z = I - J_N H_X^{-1} C_X.$$

The supplied direction-preserving inequality

$$F_Z^* G_{k,1} F_Z \preceq \Lambda_{h,n}^k H_Z$$

is retained; it is not replaced by unrelated bounds on F_Z and H_Z^+ . Use logarithmic source weights:

$$\boxed{C(\mathbf{x}) = G_\circ^{-1} + \sum_{i=1}^p e^{x_i} \Omega_i, \quad G(\mathbf{x}) = C(\mathbf{x})^{-1}, \quad \mathbf{x} \in \mathbb{R}^p.} \quad (4)$$

The weights describe the stated source penalties. The original physical joint source is the prescribed point of this family, not every point with a newly declared unchanged norm.

Fix any original flag

$$E_0 \subset E_1 \subset E.$$

Let $Q_F(\mathbf{x})$ be the full minimum metric on the specified coefficient space for E_1/E_0 , and let

$$r = \dim(E_1/E_0).$$

Put

$$R_0 = \log(5/4), \quad L_0 = \log(3/2).$$

For every real base point $\mathbf{x}_*$, the exact rational formula for Q_F has a holomorphic continuation to

$$|z_i| \leq R_0$$

around $\mathbf{x}_*$, and

$$\boxed{\left\| Q_F(\mathbf{x}_*)^{-1/2} Q_F(\mathbf{x}_* + \mathbf{z}) Q_F(\mathbf{x}_*)^{-1/2} - I \right\| \leq \frac{1}{2}.} \quad (5)$$

Its inverse satisfies

$$\boxed{\left\| Q_F(\mathbf{x}_*)^{1/2} Q_F(\mathbf{x}_* + \mathbf{z})^{-1} Q_F(\mathbf{x}_*)^{1/2} - I \right\| \leq \frac{1}{3}.} \quad (6)$$

At complex parameters these are bounds for the holomorphic continuation of the complete matrix formula. A complex parameter is not assigned a positive physical minimum.

5.1.1 1.1 Proof with the exact quotient order

Choose a fixed $G(\mathbf{x}_*)$ -orthonormal frame respecting

$$E_0 \oplus L \oplus O, \quad L = E_1 \cap E_0^{\perp G(\mathbf{x}_*)}, \quad O = E_1^{\perp G(\mathbf{x}_*)}.$$

Call its inclusion matrix F . Then

$$F^* G(\mathbf{x}_*) F = I, \quad F^{-1} = F^* G(\mathbf{x}_*).$$

In this frame,

$$D(\mathbf{z}) = F^{-1} C(\mathbf{x}_* + \mathbf{z}) F^{-*} = K_0 + \sum_i e^{z_i} K_i,$$

where

$$K_i \succeq 0, \quad \sum_{i=0}^p K_i = I.$$

The map

$$v \mapsto (K_0^{1/2} v, \dots, K_p^{1/2} v)$$

is an isometry. Therefore

$$\|D(\mathbf{z}) - I\| \leq \max_i |e^{z_i} - 1| \leq e^{R_0} - 1 = \frac{1}{4}. \quad (7)$$

Write $D = I + \Delta$. The full source metric is D^{-1} . Restrict it to E_1 , then minimize over E_0 . Block inversion gives the **inverse** of that attained quotient as

$$\boxed{E_F(\mathbf{z}) = D_{LL} - D_{LO} D_{OO}^{-1} D_{OL}.} \quad (8)$$

This follows by first inverting the restriction of D^{-1} to E_1 , and then taking its LL -block. Thus (8) is not obtained by replacing the inverse of a full matrix with an inverse of a restriction.

Equation (7) gives

$$\|E_F - I\| \leq \frac{1}{4} + \frac{(1/4)^2}{1 - 1/4} = \frac{1}{3}.$$

Hence

$$\|E_F^{-1} - I\| \leq \frac{1/3}{1 - 1/3} = \frac{1}{2}.$$

Undoing the fixed quotient-frame congruence proves (5)-(6).

The constants are independent of the spectrum of every Ω_i , of their ranks, and of whether they commute.

5.1.2 1.2 All mixed derivatives have an explicit bound

Use the logarithm determined by the convergent series for $E_F - I$. Then

$$|\log \det E_F| \leq r \log(3/2).$$

For every nonzero multi-index α , multivariable Cauchy estimates prove

$$|\partial_{\mathbf{x}}^{\alpha} \log \det Q_F(\mathbf{x})| \leq r \alpha! L_0 R_0^{-|\alpha|}. \quad (9)$$

Likewise,

$$\left\| Q_F(\mathbf{x})^{-1/2} (\partial^{\alpha} Q_F)(\mathbf{x}) Q_F(\mathbf{x})^{-1/2} \right\| \leq \frac{\alpha!}{2R_0^{|\alpha|}}. \quad (10)$$

The first derivatives also have the sharper directed identity

$$-\partial_i Q_F \succeq 0, \quad \sum_i Q_F^{-1/2} (-\partial_i Q_F) Q_F^{-1/2} \preceq I. \quad (11)$$

To verify it, differentiate the complete minimum representative. Its derivative lies in the original relation space and is orthogonal to that representative, so only the metric derivative contributes. The normalized covariance derivatives are positive matrices whose sum is at most the identity.

Thus

$$0 \leq -\sum_i \partial_i \log \det Q_F \leq r.$$

These statements include every fixed original flag quotient. They do not require a new rank theorem for a conjecturally identified subspace.

5.1.3 1.3 Finite changes have a global bound

For two real parameter values put

$$a = \min(0, y_1 - x_1, \dots, y_p - x_p),$$

$$b = \max(0, y_1 - x_1, \dots, y_p - x_p).$$

The source covariance satisfies

$$e^a C(\mathbf{x}) \preceq C(\mathbf{y}) \preceq e^b C(\mathbf{x}).$$

Taking the same full inverse, restriction and minimum proves

$$e^{-b} Q_F(\mathbf{x}) \preceq Q_F(\mathbf{y}) \preceq e^{-a} Q_F(\mathbf{x}). \quad (12)$$

The baseline covariance is the term of logarithmic weight zero; that is why zero appears in the definitions of a, b .

5.2 2. The complete projected complex current is controlled on the same domain

Retain the original observation Λ , its kernel projection $P(\mathbf{x})$, the original action A , and the same marked vector v .

At a real reference point, use its exact orthonormal kernel/observation frame. The covariance has blocks D_{KK}, D_{KB}, D_{BB} , and

$$X(\mathbf{z}) = D_{KB}D_{BB}^{-1}.$$

Equation (7) gives

$$\|X\| \leq \frac{1}{3}.$$

The complete kernel projection is

$$P(\mathbf{z}) = \begin{pmatrix} I & -X \\ 0 & 0 \end{pmatrix},$$

so

$$\|P\|, \|I - P\| \leq \frac{\sqrt{10}}{3}, \quad \|G(\mathbf{z})\| \leq \frac{4}{3}. \quad (13)$$

The physical pairings are

$$z(\mathbf{x}) = \langle Pv, A(I - P)v \rangle_{G(\mathbf{x})},$$

$$w(\mathbf{x}) = \langle (I - P)v, APv \rangle_{G(\mathbf{x})}.$$

Their holomorphic continuations are the exact formulas

$$z(\mathbf{z}) = v^* G(\mathbf{z}) P(\mathbf{z}) A(I - P(\mathbf{z})) v,$$

$$w(\mathbf{z}) = v^* G(\mathbf{z}) (I - P(\mathbf{z})) A P(\mathbf{z}) v. \quad (14)$$

For real parameters, $P^*G = GP$, so these are the original pairings.

A scalar multiple of the identity can be subtracted from A , because $P(I - P) = 0$. Put

$$\mathfrak{a} = \|A - cI\|_{G(\mathbf{x}_*)} \|v\|_{G(\mathbf{x}_*)}^2$$

for the retained arithmetic centre c . Equations (13)-(14) give

$$|z(\mathbf{z})|, |w(\mathbf{z})| \leq \frac{40}{27} \mathfrak{a}.$$

Define $f^\sharp(\mathbf{z}) = \overline{f(\bar{\mathbf{z}})}$. The actual real phase has the holomorphic continuation

$$\mathcal{P}(\mathbf{z}) = \frac{1}{2} (z^\sharp(\mathbf{z}) w(\mathbf{z}) + w^\sharp(\mathbf{z}) z(\mathbf{z})).$$

Therefore

$$\boxed{|\partial^\alpha \Re(\bar{z} w)(\mathbf{x}_*)| \leq \left(\frac{40}{27}\right)^2 \mathfrak{a}^2 \alpha! R_0^{-|\alpha|}.} \quad (15)$$

Both moving projections and both complex products remain. Equation (15) bounds their variation; it does not assign their sign.

It also yields a finite Taylor certificate. On

$$\|\mathbf{z}\|_1 \leq \theta R_0, \quad 0 < \theta < 1,$$

the total-degree- N Taylor remainder is at most

$$\boxed{\left(\frac{40}{27}\right)^2 a^2 \frac{\theta^{N+1}}{1-\theta}}. \quad (16)$$

The same argument gives $rL_0\theta^{N+1}/(1-\theta)$ for the logarithmic determinant.

All derivatives are calculable through finite matrix recurrences. For example, differentiating $C G = I$ gives

$$\boxed{G_\alpha = -G_0 \sum_{0 < \beta \leq \alpha} \binom{\alpha}{\beta} C_\beta G_{\alpha-\beta}}. \quad (17)$$

The order of the factors is part of the formula. No commutation of the source additions is used.

5.3 3. Heat control is uniform in the heat time

The preceding derivative bounds concern matrices and phases. The full positive heat operator admits an additional control independent of the largest native action norm.

Let M be the fixed original centred arithmetic action. Put

$$H(\mathbf{x}) = M^{\dagger G(\mathbf{x})} M.$$

For two positive metrics in (12), singular-value comparison through the coefficient identity gives

$$e^{-(b-a)/2} s_j(M; G(\mathbf{x})) \leq s_j(M; G(\mathbf{y})) \leq e^{(b-a)/2} s_j(M; G(\mathbf{x})).$$

Since

$$\sup_{u \geq 0} u e^{-u} = 1/e,$$

we obtain

$$\boxed{\sup_{\tau \geq 0} \left| \text{Tr} e^{-\tau H(\mathbf{y})} - \text{Tr} e^{-\tau H(\mathbf{x})} \right| \leq \frac{\text{rank } M}{e} (b-a)}. \quad (18)$$

Likewise, for every $a_* > 0$,

$$\boxed{|\log \det(a_* I + H(\mathbf{y})) - \log \det(a_* I + H(\mathbf{x}))| \leq \text{rank } M (b-a)}. \quad (19)$$

Zero singular directions contribute the same constant at both endpoints.

5.3.1 3.1 The actual projected full heat has a derivative bound too

Consider a straight logarithmic path

$$\mathbf{x}(s) = \mathbf{x}_0 + s\mathbf{V},$$

and put

$$\Delta_{\mathbf{v}} = \max(0, v_1, \dots, v_p) - \min(0, v_1, \dots, v_p).$$

Use the explicitly metric-compatible moving frame

$$F'(s) = -\frac{1}{2}G^{-1}G'F.$$

In that frame,

$$F^*GF = I, \quad B = F^*G'F = B^*,$$

and the original fixed arithmetic action satisfies

$$\dot{M} = \frac{1}{2}[B, M].$$

The spectrum of B has width at most $\Delta_{\mathbf{v}}$. Its scalar midpoint can be subtracted from the commutator.

For $H = M^*M$,

$$\dot{H} = M^*BM - \frac{1}{2}(BH + HB).$$

Writing $M = UH^{1/2}$, with the actual polar factor, and applying the divided-difference formula for $e^{-\tau H}$, gives

$$\|\nabla_s e^{-\tau H}\|_{\text{HS}} \leq \left(\frac{1}{e} + \frac{1}{2}\right) \frac{\sqrt{q}}{2} \Delta_{\mathbf{v}}, \quad \tau \geq 0. \quad (20)$$

The two scalar estimates behind this are

$$\begin{aligned} \sqrt{xy} \left| \frac{e^{-\tau x} - e^{-\tau y}}{x - y} \right| &\leq \frac{1}{e}, \\ \frac{x + y}{2} \left| \frac{e^{-\tau x} - e^{-\tau y}}{x - y} \right| &\leq \frac{1}{2}. \end{aligned} \quad (21)$$

The first follows by differentiating in $\log x$ and using that the logarithmic mean is at least the geometric mean. For the second, write $\tau x = b + d, \tau y = b$. The quantity before division by two is

$$e^{-b} \left(1 + \frac{2b}{d}\right) (1 - e^{-d}).$$

Its maximum over $b \geq 0$ is at $b = 0$ for $d \geq 2$, and at $b = 1 - d/2$ for $d < 2$; both maxima are below one. Repeated and zero eigenvalues are covered by the continuous limits.

For the original fixed kernel, the projection derivative in this frame is

$$\nabla_s P_K = \frac{1}{2}[(I - P_K)BP_K + P_KB(I - P_K)]. \quad (22)$$

If P is this projection or its complement, of rank ℓ , its positive derivative trace is at most

$$\frac{1}{4} \min(\ell, q - \ell) \Delta_{\mathbf{v}}.$$

Combining (20)–(22),

$$\left| \frac{d}{ds} \operatorname{Tr} \left(P(s) e^{-\tau H(s)} \right) \right| \leq \Delta_{\mathbf{v}} \left[\frac{\min(\ell, q - \ell)}{4} + \frac{\frac{1}{2} + \frac{1}{e}}{2} \sqrt{\ell q} \right]. \quad (23)$$

For the original observation,

$$\operatorname{Tr}_B(\Lambda e^{-\tau H} L) = \operatorname{Tr}_E((I - P_K) e^{-\tau H}).$$

Thus (23) is a bound for the actual observed full heat, not a trace in a different observation space.

A varying period can also move Λ . Its additional frame derivative must then be retained. No dimension-only bound can control an arbitrarily fast reparameterization of that extra motion.

5.4 4. Every multiscale covariance family has an explicit metric normal form

The bounds above hold at all positive source weights. The following calculation determines the complete boundary scales.

After the fixed initial isometry, consider

$$C(T) = I + \sum_{j=1}^s T^{a_j} K_j, \quad a_1 > \cdots > a_s > 0, \quad K_j \succeq 0. \quad (24)$$

Equal weights have already been combined by addition of their actual covariance matrices.

Set

$$V_j = \sum_{i \leq j} \operatorname{ran} K_i, \quad E_j = V_j \cap V_{j-1}^\perp,$$

and let P_j project onto E_j . Let P_0 project onto $V_s^\perp$.

Define

$$\begin{aligned} D_T &= P_0 + \sum_j T^{a_j/2} P_j, \\ H_j &= P_j K_j P_j|_{E_j}, \quad H_\infty = P_0 + \sum_j P_j K_j P_j. \end{aligned} \quad (25)$$

Every nonempty H_j is positive definite. Indeed, a vector in E_j annihilated by K_j is orthogonal both to V_{j-1} and to $\operatorname{ran} K_j$, hence to V_j , and is zero.

The exact limit is

$$\boxed{D_T^{-1} C(T) D_T^{-1} \longrightarrow H_\infty.} \quad (26)$$

5.4.1 4.1 A finite error, not merely a limit

For $j > 1$, set

$$\delta_j(T) = T^{-(a_{j-1}-a_j)/2}.$$

Then

$$\left\| D_T^{-1} C(T) D_T^{-1} - H_\infty \right\| \leq T^{-a_s} + \sum_{j=2}^s (2\delta_j + \delta_j^2) \|K_j\|. \quad (27)$$

To prove it, the j -th normalized summand is

$$(P_j + R_j(T)) K_j (P_j + R_j(T)),$$

where $R_j(T)$ acts only on V_{j-1} and has norm at most $\delta_j(T)$. The identity contribution supplies T^{-a_s} .

There is also an all- $T \geq 1$ comparison with no division by the exponent gaps.

Put

$$W_j = K_j P_j H_j^{-1}, \quad L = \sum_j \|W_j - P_j\|, \quad \gamma = \min(1, \min_j \lambda_{\min} H_j).$$

The exact factor

$$K_j \succeq W_j H_j W_j^*$$

follows by inserting the orthogonal projection onto $\text{ran}(K_j^{1/2} P_j)$.

The matrix

$$U = P_0 + \sum_j W_j P_j$$

is block upper triangular with identity diagonal. After conjugation by D_T , its strictly upper part has norm at most L , and its inverse is the finite nilpotent series. Therefore

$$\frac{\gamma}{(1 + L + \dots + L^s)^2} I \preceq D_T^{-1} C(T) D_T^{-1} \preceq \left(1 + \sum_j \|K_j\| \right) I. \quad (28)$$

This remains valid when several source ranges coincide or some layer is empty. For arbitrary positive weights, sort them together with the baseline weight one. The same construction applies on each of the finitely many ordering charts; sub-unit weights enter the upper bound, while the baseline controls their remaining directions.

5.4.2 4.2 All singular and exterior scales of the arithmetic action

The exact model metric is

$$G_{\text{mod}}(T) = D_T^{-1} H_\infty^{-1} D_T^{-1}.$$

Its isometry to Euclidean coordinates is

$$S_T = H_\infty^{-1/2} D_T^{-1}.$$

Choose orthonormal bases inside the layers. Let their weights be $w_1, \dots, w_q$, including zero on $V_s^\perp$, and put

$$\hat{A} = H_\infty^{-1/2} A H_\infty^{1/2}.$$

Then

$$(S_T A S_T^{-1})_{ij} = T^{(w_j - w_i)/2} \hat{A}_{ij}.$$

For every exterior rank $r \leq \text{rank } A$, define

$$\beta_r = \max_{\det \hat{A}_{I,J} \neq 0} \frac{\sum_{j \in J} w_j - \sum_{i \in I} w_i}{2}. \quad (29)$$

Let $\mathcal{A}_r$ be the full exterior matrix retaining precisely the minors attaining that exponent. It is nonzero. Then

$$T^{-\beta_r} \|\wedge^r A\|_{G(T)} \longrightarrow \|\mathcal{A}_r\|. \quad (30)$$

The finite relative comparison in (27) bounds the error before taking a limit. In particular, it supplies an actual guard when the coefficient data themselves vary with k ; a fixed-data $o(1)$ has not been declared uniform in k .

For operator norm,

$$\beta_1 = \max_{P_i A P_j \neq 0} \frac{a_j - a_i}{2},$$

with the complementary weight zero included. Thus

$$\|A\|_{G(T)} \text{ stays bounded} \iff AV_j \subset V_j \text{ for every } j. \quad (31)$$

The equivalence follows directly from the displayed action blocks. It does not assume an arithmetic invariant filtration.

For an invertible A , the full exterior exponent is zero:

$$\det(S_T A S_T^{-1}) = \det A.$$

Growing and shrinking singular directions therefore compensate exactly.

5.4.3 4.3 Complete fixed subquotients

For a fixed inclusion $B : \mathbb{C}^m \rightarrow E$, set

$$C_B = H_\infty^{-1/2} B.$$

The r -th exterior norm of $D_T^{-1} C_B$ is governed by the least sum of row weights among its nonzero $r \times r$ minors. Denote that sum by w_r , with $w_0 = 0$, and the norm of the full leading exterior matrix by $c_r > 0$.

Then the eigenvalues of the actual restricted metric satisfy

$$\lambda_r(B^* G(T) B) \sim \left(\frac{c_r}{c_{r-1}} \right)^2 T^{-(w_r - w_{r-1})}. \quad (32)$$

The matrix of all leading minors is retained, not a single favoured minor.

For a quotient E/E_0 , its exact covariance is

$$\Lambda_0 C(T) \Lambda_0^*.$$

For E_1/E_0 , restrict that attained quotient to its actual image of E_1 . Every representative of that value lies in E_1 , because its difference from a chosen lift lies in E_0 . Thus this order of quotient and restriction is exactly the original full minimum.

Applying (24)–(32) to the projected source matrices therefore calculates every fixed flag’s metric and exterior scales.

5.4.4 4.4 A noncommuting check in closed form

Take

$$b_1 = (1, 1, 0)^T, \quad b_2 = (1, i, 1)^T, \quad b_3 = (1, 0, 0)^T,$$

and

$$C(T) = I + T^3 b_1 b_1^* + T b_2 b_2^* + T^2 b_3 b_3^*.$$

The three layers are

$$\frac{(1, 1, 0)}{\sqrt{2}}, \quad \frac{(1, -1, 0)}{\sqrt{2}}, \quad (0, 0, 1),$$

with weights 3, 2, 1, and

$$H_\infty = \text{diag}(2, 1/2, 1).$$

Hence the covariance eigenvalues are asymptotic to

$$2T^3, \quad \frac{1}{2}T^2, \quad T.$$

For the original-coordinate subspace $\mathbb{C}e_1$,

$$e_1^* C(T)^{-1} e_1 \sim T^{-2}.$$

For $A = e_3 e_1^*$,

$$\|A\|_{C(T)^{-1}} \sim T.$$

This agrees with the exact layer calculation (29). No commuting approximation was used.

5.5 5. The full observed memory has a gap-free positive representation

Return to the original metric and its orthogonal decomposition

$$E = K \oplus LB.$$

Write the original arithmetic operator as

$$M = \begin{pmatrix} M_K & D \\ C & M_B \end{pmatrix}.$$

Then

$$H = M^\dagger M = \begin{pmatrix} A & B_H \\ B_H^\dagger & D_H \end{pmatrix},$$

where

$$A = M_K^\dagger M_K + C^\dagger C,$$

$$B_H = M_K^\dagger D + C^\dagger M_B,$$

$$D_H = D^\dagger D + M_B^\dagger M_B. \quad (33)$$

These are the complete original blocks in the repository's observed-heat equation. In particular, D_H is not replaced by $M_B^\dagger M_B$.

Positivity proves

$$\ker A \subset \ker B_H^\dagger, \quad S = D_H - B_H^\dagger A^+ B_H \succeq 0.$$

Let P_λ be the spectral projections of A for $\lambda > 0$, and define

$$C_\lambda = \lambda^{-1} B_H^\dagger P_\lambda B_H \succeq 0.$$

Then

$$\sum_{\lambda>0} C_\lambda = B_H^\dagger A^+ B_H =: T \preceq D_H. \quad (34)$$

The memory kernel and its Laplace-domain pencil are exactly

$$\boxed{\mathcal{M}(\tau) = B_H^\dagger e^{-\tau A} B_H = \sum_{\lambda>0} \lambda e^{-\lambda \tau} C_\lambda,} \quad (35)$$

$$\boxed{\mathcal{F}(z) = zI + D_H - B_H^\dagger (zI + A)^{-1} B_H = zI + S + \sum_{\lambda>0} \frac{z}{z + \lambda} C_\lambda.} \quad (36)$$

The compressed resolvent is

$$\boxed{\Lambda(zI + H)^{-1} L = \mathcal{F}(z)^{-1}.} \quad (37)$$

Every C_λ is a full positive matrix; they need not commute.

5.5.1 5.1 All memory derivatives have a common integral bound

For every integer $n \geq 0$,

$$(-1)^n \mathcal{M}^{(n)}(\tau) \succeq 0,$$

and

$$\boxed{\int_0^\infty \tau^n (-1)^n \mathcal{M}^{(n)}(\tau) d\tau = n! T \preceq n! D_H.} \quad (38)$$

Pointwise,

$$\boxed{\tau^{n+1} (-1)^n \mathcal{M}^{(n)}(\tau) \preceq \left(\frac{n+1}{e} \right)^{n+1} T.} \quad (39)$$

These follow by integrating or maximizing the actual scalar factors $\lambda^{n+1}\tau^n e^{-\lambda\tau}$.

There is no lower bound on the positive λ in (38)–(39). Long memory at small λ remains present with its exact weight.

For $z > 0$,

$$\mathcal{F}'(z) = I + \sum_{\lambda} \frac{\lambda}{(z + \lambda)^2} C_{\lambda}.$$

For all $n \geq 1$, subtracting I at $n = 1$,

$$0 \preceq z^n (-1)^{n-1} (\mathcal{F}^{(n)}(z) - \mathbf{1}_{n=1} I) \preceq n! \frac{n^n}{(n+1)^{n+1}} T. \quad (40)$$

For $\Im z > 0$,

$$\Im \mathcal{F}(z) = (\Im z) \left[I + \sum_{\lambda} \frac{\lambda}{|z + \lambda|^2} C_{\lambda} \right] \succ 0. \quad (41)$$

Thus the analytic pencil is controlled on the full upper half-plane, not only at positive real regularizers.

5.5.2 5.2 The exact smallest memory realization

Let

$$r_{\text{mem}} = \sum_{\lambda > 0} \text{rank } C_{\lambda}.$$

The coupled kernel space is

$$K_{\text{coupled}} = \bigoplus_{\lambda > 0} \text{ran}(P_{\lambda} B_H).$$

Its orthogonal complement is invariant under A and annihilated by $B_H^{\dagger}$.
On

$$\bigoplus_{\lambda > 0} \text{ran } C_{\lambda},$$

define

$$A_{\min} = \bigoplus_{\lambda} \lambda I, \quad B_{\min} = \text{stack}_{\lambda} \sqrt{\lambda} C_{\lambda}^{1/2}.$$

The explicit isometry to the original coupled kernel is

$$J_{\lambda} = \lambda^{-1/2} P_{\lambda} B_H C_{\lambda}^{+1/2}. \quad (42)$$

It intertwines both $A_{\min}$ and $B_{\min}$ with the original blocks.
The reduced positive matrix

$$\begin{pmatrix} A_{\min} & B_{\min} \\ B_{\min}^{\dagger} & D_H \end{pmatrix}$$

has the exact same observed resolvent and heat as (33). The discarded original kernel summand remains as a separate decoupled block in the full trace.

This realization is minimal among positive selfadjoint finite realizations of the same memory: the residue at each distinct pole $-\lambda$ has rank $\text{rank } C_\lambda$, requiring at least that many states at that eigenvalue.

5.6 6. Complete heat and determinant mixing are bounded by the coupling rank

Put

$$H_0 = \text{diag}(A, D_H), \quad r_c = \text{rank } B_H.$$

Define the complete heat-mixing quantity

$$\mathcal{Q}(\tau) = \text{Tr } e^{-\tau H} - \text{Tr } e^{-\tau A} - \text{Tr } e^{-\tau D_H}.$$

Then

$$0 \leq \mathcal{Q}(\tau) \leq \min\{r_c, \tau^2 \|B_H\|_{\text{HS}}^2\}. \quad (43)$$

5.6.1 6.1 Positivity retains the complete operator

Choose eigenbases of A and D_H . For each corresponding unit vector v , the spectral measure of the full H and scalar convexity give

$$\langle v, e^{-\tau H} v \rangle \geq e^{-\tau \langle v, H v \rangle}.$$

Sum over the two bases. This proves $\mathcal{Q}(\tau) \geq 0$.

This is a trace inequality. It does not assert the generally unjustified operator inequality

$$P_B e^{-\tau H} P_B \succeq e^{-\tau D_H}.$$

5.6.2 6.2 The rank bound

The off-diagonal perturbation

$$O = \begin{pmatrix} 0 & B_H \\ B_H^\dagger & 0 \end{pmatrix}$$

has positive and negative parts of rank r_c . First add O_+ to H_0 , then subtract O_- .

Min-max shows that adding O_+ cannot increase the trace of the decreasing scalar function $e^{-\tau x}$. Subtracting a rank- r_c positive matrix can increase that trace by at most r_c , because the scalar function lies between zero and one. This proves the first upper bound.

5.6.3 6.3 The quadratic bound

The path

$$H_s = H_0 + sO, \quad 0 \leq s \leq 1,$$

is positive. For

$$\phi(s) = \text{Tr} e^{-\tau H_s},$$

one has $\phi'(0) = 0$, and

$$\phi''(s) = \tau^2 \int_0^1 \text{Tr} \left(O e^{-(1-u)\tau H_s} O e^{-u\tau H_s} \right) du.$$

In the actual eigenbasis of H_s , this is a sum of

$$|O_{ij}|^2 e^{-\tau[(1-u)\lambda_i + u\lambda_j]}.$$

It is nonnegative and at most $\tau^2 \|O\|_{\text{HS}}^2$. Since

$$\|O\|_{\text{HS}}^2 = 2\|B_H\|_{\text{HS}}^2,$$

integration against $1 - s$ proves the second upper bound in (43).

5.6.4 6.4 Return to the original observed compression

The same scalar Jensen calculation on the observed block gives

$$0 \leq \text{Tr}(\Lambda e^{-\tau H} L) - \text{Tr} e^{-\tau D_H} \leq \mathcal{Q}(\tau). \quad (44)$$

Since

$$D_H = M_B^\dagger M_B + D^\dagger D,$$

min-max and a positive interpolation give

$$0 \leq \text{Tr} e^{-\tau M_B^\dagger M_B} - \text{Tr} e^{-\tau D_H} \leq \min\{\text{rank } D, \tau \|D\|_{\text{HS}}^2\}.$$

Combining,

$$\boxed{-\min\{\text{rank } D, \tau \|D\|_{\text{HS}}^2\} \leq \text{Tr}(\Lambda e^{-\tau H} L) - \text{Tr} e^{-\tau M_B^\dagger M_B} \leq \min\{\text{rank } B_H, \tau^2 \|B_H\|_{\text{HS}}^2\}.} \quad (45)$$

Both ranks are at most $\dim K$. This proves (1)-(2).

The error can be order k on the original simple family, so it is not automatically negligible at the arithmetic k -scale. It is uniformly negligible after normalization by $q = (k+1)^2$.

5.6.5 6.5 The complete logarithmic mixing is also explicit

Define

$$g(z) = \log \det(zI + A) + \log \det(zI + D_H) - \log \det(zI + H).$$

The full determinant identity and (43) give

$$\boxed{g(z) = \log \det(zI + D_H) - \log \det \mathcal{F}(z) = \int_0^\infty e^{-z\tau} \frac{\mathcal{Q}(\tau)}{\tau} d\tau.} \quad (46)$$

Thus g is completely monotone on $z > 0$, and

$$\boxed{0 \leq g(z) \leq \frac{\|B_H\|_{\text{HS}}^2}{z^2},}$$

$$0 \leq (-1)^n g^{(n)}(z) \leq \min \left\{ \frac{r_c(n-1)!}{z^n}, \frac{\|B_H\|_{\text{HS}}^2(n+1)!}{z^{n+2}} \right\}, \quad n \geq 1. \quad (47)$$

A useful rank form is

$$g(z) \leq r_c \log \left(1 + \frac{\min(\|A\|, \|D_H\|)}{z} \right). \quad (48)$$

It follows by applying the same Schur determinant on either block and using $\mathcal{F}(z) \succeq zI$.

The local leakage correction is

$$\ell(z) = \log \det(zI + D_H) - \log \det(zI + M_B^\dagger M_B) \geq 0.$$

The actual observed pencil therefore satisfies

$$\log \det \mathcal{F}(z) - \log \det(zI + M_B^\dagger M_B) = \ell(z) - g(z). \quad (49)$$

Both nonnegative terms remain; their difference is not assigned a sign.

At every original cutoff,

$$|\ell(z) - g(z)| \leq m \log \left(1 + \frac{\|M\|_G^2}{z} \right). \quad (50)$$

For the supplied joint-source metric,

$$\log^+ \|M\|_{G_N^J} = O_h(k \log^2(q+2)).$$

Hence at fixed $z > 0$, the right side of (50) is

$$O_h(k^2 \log^2(q+2)) = o_h(kq). \quad (51)$$

This completes the joint-source observed positive spectral comparison at that scale. The corresponding canonical bound can still be of order kq , and has not been assigned the joint value. The original source explicitly distinguishes those metrics and their leakage term.

5.7 7. The conductor detects every sufficiently low-degree proper rational family

This extends the repository's original inverse-power calculation to arbitrary poles, their full multiplicities, and collisions among translated poles.

Retain the original finite translation operator

$$\mathcal{T}_A f(S) = \sum_{j=1}^J a_j f(S + b_j),$$

with nonzero symbol

$$E_A(z) = \sum_j a_j e^{b_j z}, \quad v = \text{ord}_0 E_A, \quad \mu_v \neq 0.$$

Coincident shifts can be combined by their exact coefficient sum. The labelled product below may still retain all of them.

Let χ be the original upper relation and $\hat{\chi}$ the lower one. The induced full-jet map is justified by the exact divisibilities

$$\hat{\chi}(S) \mid \chi(S + b_j). \quad (52)$$

The original map $C_A : E \rightarrow E_-$ therefore satisfies

$$C_A[P] = [\mathcal{T}_A P], \quad K \subset \ker C_A.$$

Choose any monic polynomial

$$D(S) = \prod_{\alpha} (S - z_{\alpha})^{m_{\alpha}}, \quad d = \deg D, \quad (D, \chi) = 1.$$

Define the original arithmetic subspace

$$\boxed{U_D p = [p/D], \quad \deg p < d.} \quad (53)$$

The physical value remains $\eta U_D p$, with the original full unit in η .

5.7.1 7.1 A nonzero translation operator has no proper-rational kernel

For a nonzero proper rational f , write its actual expansion at infinity

$$f(S) = cS^{-n_0} + O(S^{-n_0-1}), \quad n_0 \geq 1, \quad c \neq 0.$$

The first nonzero term of its translated sum is

$$\boxed{\mathcal{T}_A f(S) = (-1)^v \binom{n_0 + v - 1}{v} \mu_v c S^{-n_0-v} + O(S^{-n_0-v-1}).} \quad (54)$$

All contributions from higher Laurent coefficients have a symbol moment of order below v at this leading power and vanish. The displayed coefficient is nonzero.

Thus

$$\boxed{\mathcal{T}_A f = 0 \implies f = 0} \quad (55)$$

on the complete space of proper rational functions.

This includes translated-pole collisions. It is not a simple-pole argument.

5.7.2 7.2 The exact finite observation guard

Use the fixed-degree common denominator

$$\mathcal{B}(S) = \prod_{j=1}^J D(S + b_j), \quad L = Jd.$$

For $f = p/D$,

$$N_p(S) = \mathcal{B}(S) \mathcal{T}_A(p/D)(S)$$

is a polynomial. Equation (54) gives

$$\deg N_p \leq L - v - 1.$$

The denominator is coprime to $\hat{\chi}$, by (52) and $(D, \chi) = 1$. Therefore vanishing of all original lower jets makes $\hat{\chi}$ divide N_p .

On the explicit degree domain

$$\boxed{Jd - v \leq q' := \deg \hat{\chi}}, \quad (56)$$

this forces $N_p = 0$, then $p = 0$ by (55). Hence

$$\boxed{\ker C_A \cap U_D \mathcal{P}_{d-1} = 0}. \quad (57)$$

The same proof uses the smaller degree of the actual least common denominator when some translated poles coincide. For a family crossing such collisions, the fixed product $\mathcal{B}$ gives one constant-degree formula.

Equation (57), together with $K \subset \ker C_A$, proves (3). The complexity is always finite: $D = \chi + 1$ and the original remainder polynomial give a proper rational representative of degree at most q .

5.7.3 7.3 A global coefficient inverse survives changes of the symbol order

Define the polynomial columns

$$F_l(S) = \mathcal{B}(S) \sum_{j=1}^J a_j \frac{(S + b_j)^l}{D(S + b_j)}, \quad 0 \leq l < d.$$

Let $\mathcal{T}_D$ be their coefficient matrix through degree $L - 1$.

It has rank d whenever the original symbol is nonzero, by (55). Its entries are polynomial expressions in the coefficients of D , the shifts, and the conductor coefficients. No difference between pole centres is inverted.

On the uniform family domain $L \leq q'$, take the actual lower quotient value $y = C_A U_D p$. Then

$$N_p = \text{rem}_{\hat{\chi}}(\mathcal{B}y)$$

as an actual polynomial of degree below L . Thus

$$\boxed{W_D = (\mathcal{T}_D^* \mathcal{T}_D)^{-1} \mathcal{T}_D^* \text{coeff}_{<L} M_{\mathcal{B}}^{(-)}} \quad (58)$$

satisfies

$$\boxed{W_D C_A U_D = I_d}. \quad (59)$$

Here $M_{\mathcal{B}}^{(-)}$ is multiplication by $\mathcal{B}$ in the original lower quotient. All complete lower jets are converted by their actual Hermite map.

This inverse is real analytic across a zero of μ_0 , or any other change in the first nonzero symbol moment, as long as the full operator remains nonzero. Its finite norm has the bound

$$\boxed{\|W_D\| \leq \left[\frac{\text{Tr}(\mathcal{T}_D^* \mathcal{T}_D)^{d-1}}{\det(\mathcal{T}_D^* \mathcal{T}_D)} \right]^{1/2} \left\| \text{coeff}_{<L} M_{\mathcal{B}}^{(-)} \right\|}. \quad (60)$$

The Gram in this expression is a specified finite polynomial coefficient Gram, not an unevaluated native observation kernel.

For example,

$$\mathcal{T}_\epsilon f(S) = (1 + \epsilon)f(S + 1) - f(S), \quad D(S) = S,$$

has

$$\mathcal{B}(S)\mathcal{T}_\epsilon(1/S) = \epsilon S - 1.$$

Its coefficient column is $(-1, \epsilon)^T$, and its inverse uses

$$\frac{(-1, \bar{\epsilon})}{1 + |\epsilon|^2}.$$

It remains bounded at $\epsilon = 0$, where the symbol order changes from zero to one. Dividing only by $\mu_0 = \epsilon$ would have introduced an artificial singularity.

5.7.4 7.4 The original symbol also gives a triangular inverse

Write

$$f(S) = \sum_{n \geq 1} c_n S^{-n}, \quad \mathcal{T}_A f(S) = \sum_{m \geq 1} b_m S^{-m},$$

and

$$e(z) = z^{-v} E_A(-z), \quad h_j = [z^j] e(z)^{-1}.$$

Then

$$c_n = (n-1)! \sum_{l=1}^n h_{n-l} \frac{b_{v+l}}{(v+l-1)!}, \quad 1 \leq n \leq d. \quad (61)$$

Finally,

$$p(S) = \text{pol}_{S=\infty} \left[D(S) \sum_{n=1}^d c_n S^{-n} \right]. \quad (62)$$

To verify (61), the coefficient of S^{-m} is

$$b_m = \sum_{n=1}^m (-1)^{m-n} \binom{m-1}{m-n} \mu_{m-n} c_n.$$

Division by $(m-1)!$, at $m = v+l$, turns this into ordinary finite convolution by $e(z)$. Equations (61)–(62) invert that convolution.

The two inverse constructions agree on the actual conductor image because both recover p . They need not agree on unrelated output vectors.

5.7.5 7.5 Pole collisions retain the full arithmetic metric

In separated partial-fraction coordinates, the numerator columns are

$$D(S)/(S - z_\alpha)^j.$$

Their coefficient matrix $\mathcal{P}_D$, ordered by root and increasing pole order, has

$$\det \mathcal{P}_D = (-1)^{d(d-1)/2} \prod_{\alpha < \beta} (z_\beta - z_\alpha)^{m_\alpha m_\beta}. \quad (63)$$

This follows from the full Hermite evaluation map and the local triangular division by $D/(S - z_\alpha)^{m_\alpha}$.

The numerator coordinate p remains a fixed d -dimensional frame through collisions. For any original positive arithmetic metric,

$$\boxed{G_{\text{partial}} = \mathcal{P}_D^*(U_D^* G U_D) \mathcal{P}_D.} \quad (64)$$

The observed Gram has exactly the same congruence. Thus the determinant vanishing of the separated-pole frame is retained, while the full confluent numerator frame remains nondegenerate.

For instance,

$$\frac{1}{\epsilon} \left(\frac{1}{S - \epsilon} - \frac{1}{S} \right) \rightarrow \frac{1}{S^2}.$$

The numerator of the left side in the common denominator $S(S - \epsilon)$ is exactly one. This is the explicit recovery of the additional confluent pole, not deletion of the diverging separated coefficients.

5.7.6 7.6 Return through the actual native observation norm

The original map has

$$C_A = R_A \Lambda, \quad R_A = C_A L_N.$$

For

$$Q_N = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad \Gamma_A = C_A G_N^{-1} C_A^*,$$

one has

$$R_A Q_N^{-1} R_A^* = \Gamma_A.$$

The complete lower minimum therefore gives

$$\boxed{U_D^* \Lambda^* Q_N \Lambda U_D \succeq (C_A U_D)^* \Gamma_A^{-1} (C_A U_D).} \quad (65)$$

With $H_D = U_D^* G_N U_D$, equations (59) and (65) give

$$\boxed{U_D^* \Lambda^* Q_N \Lambda U_D \succeq \ell_D H_D, \quad \ell_D = \frac{1}{\lambda_{\max} \Gamma_A \|W_D\|^2 \lambda_{\max} H_D} > 0.} \quad (66)$$

Every quantity is in the original metrics. In particular, the full arithmetic unit remains in Λ and G_N .

5.8 8. Multipole detection controls the full slow singular subspace of an arithmetic filter

Let $p(y)$ be monic of degree d , coprime to the original relation. Retain the original full minimum polynomial section

$$\widehat{R}_N : E \rightarrow \mathcal{P}_N.$$

Divide its actual polynomial representative:

$$\widehat{R}_N x = p \mathfrak{Q}_p \widehat{R}_N x + \mathfrak{R}_p \widehat{R}_N x.$$

Then

$$\boxed{p(M)^{-1} = B_p + U_p T_p,} \quad (67)$$

where

$$B_p = J_N \mathfrak{Q}_p \widehat{R}_N, \quad T_p = \mathfrak{R}_p \widehat{R}_N, \quad U_p r = [r/p].$$

The first term has the finite original bound

$$\|B_p\| \leq D_p := \|\mathfrak{Q}_p\|_{H_N \rightarrow H_{N-d}}. \quad (68)$$

Its coefficient matrix is the actual long-division map, and its norm uses the full native moment matrices.

Put $X = p(M)^{-1}$, and order its singular values decreasingly:

$$\sigma_1(X) \geq \cdots \geq \sigma_q(X) > 0.$$

Let V_d be its top d left singular subspace. Since the second term of (67) has range in $\text{im } U_p$ and rank at most d ,

$$\sigma_{d+1}(X) \leq D_p,$$

and

$$\boxed{\|(I - P_{U_p})P_{V_d}\| \leq \varepsilon_p := \min \left\{ 1, \frac{D_p}{\sigma_d(X)} \right\}.} \quad (69)$$

Apply the native rational observation bound ℓ_p from (66). Every unit vector in V_d then has observed fraction at least

$$\boxed{\ell_{\text{slow}} = \left(\sqrt{\ell_p} \sqrt{1 - \varepsilon_p^2} - \varepsilon_p \right)_+^2.} \quad (70)$$

The left singular vectors of $p(M)^{-1}$ are the right singular vectors of $p(M)$. Therefore the full filtered heat satisfies

$$\boxed{\text{Tr} \left(P_B e^{-\tau p(M)^\dagger p(M)} \right) \geq \ell_{\text{slow}} \sum_{j=1}^d e^{-\tau/\sigma_j(X)^2}.} \quad (71)$$

Its kernel counterpart has the complete upper bound

$$\boxed{\text{Tr} \left(P_K e^{-\tau p(M)^\dagger p(M)} \right) \leq (1 - \ell_{\text{slow}}) \sum_{j=1}^d e^{-\tau/\sigma_j(X)^2} + (q - d) e^{-\tau/D_p^2}.} \quad (72)$$

The last term is absent at $d = q$.

This generalizes the source's one inverse-power late-heat comparison to the entire proper rational family. The small matrix

$$H_p^{1/2} T_p G_N^{-1} T_p^* H_p^{1/2}$$

contains the nonzero singular values of the finite-rank part $U_p T_p$, so the finite guard in (69) can also be evaluated using that matrix and D_p .

The right side of (70) may be zero when the available approximation is not sufficiently accurate. It is not replaced by a positive constant without checking the stated finite quantities.

5.9 9. Rank-changing parameter families have an explicit boundary correction

Uniform logarithmic source-weight control does not justify continuity of a literal source image when that image loses rank. The complete extension can nevertheless be constructed.

Let $B(p)$ be a holomorphic finite source coefficient matrix in a fixed ambient source with positive metric $H(p)$. Include the original scalar source X among its columns, with the original value map $J|_X$ onto E .

On the maximal-rank locus, consider

$$p \mapsto \text{im } B(p)$$

in the Grassmannian. Take the closure of its graph. Every limiting plane Y contains both the literal image at that point and the original X .

On a Grassmann chart,

$$Y = \text{im} \begin{pmatrix} I \\ Z \end{pmatrix}.$$

Its full source Gram and value map are

$$H_Y = \begin{pmatrix} I \\ Z \end{pmatrix}^* H \begin{pmatrix} I \\ Z \end{pmatrix}, \quad J_Y = J \begin{pmatrix} I \\ Z \end{pmatrix},$$

with the obvious parenthesized interpretation $H_Y = [I; Z]^* H [I; Z]$. The attained quotient is

$$\boxed{Q_Y = (J_Y H_Y^{-1} J_Y^*)^{-1}}. \quad (73)$$

It stays positive because Y contains X , and

$$\boxed{Q_{\text{ambient}} \preceq Q_Y \preceq Q_X}. \quad (74)$$

At a rank-drop point, orthogonalize the extra columns of Y against the **whole literal source**. If their actual normal Gram and value map are $H_{\text{miss}}, F_{\text{miss}}$, then

$$\boxed{Q_Y = [Q_{\text{literal}}^{-1} + F_{\text{miss}} H_{\text{miss}}^+ F_{\text{miss}}^*]^{-1}}. \quad (75)$$

This is the boundary correction, with its complete minimum.

5.9.1 9.1 A two-parameter example displays the exceptional fibre

In $\mathbb{C}^3$, with its Euclidean metric and

$$J(x_1, x_2, x_3) = x_1 + x_2 + x_3,$$

retain $X = \mathbb{C}e_1$ and

$$R_{z_1, z_2} = (1 - z_1 - z_2)e_1 + z_1e_2 + z_2e_3.$$

It has $JR = 1$. The full normal data are

$$H_Z = |z_1|^2 + |z_2|^2, \quad F_Z = z_1 + z_2,$$

so

$$|F_Z|^2 \leq 2H_Z$$

everywhere.

Along the direction $[a : b]$, the limiting joint metric is

$$\boxed{\frac{|a|^2 + |b|^2}{|a|^2 + |b|^2 + |a + b|^2}}. \quad (76)$$

It equals $1/3$, $1/2$, or 1 on the directions $[1 : 1]$, $[1 : 0]$, and $[1 : -1]$. The literal metric at the origin is one.

The graph extension retains the projective direction $[a : b]$. It does not select one of these different limits and call the original metric continuous.

5.9.2 9.2 Observation-rank loss has a separate factor

For a holomorphic observation of generic rank r , use a row-Grassmann chart

$$\Lambda_{\text{sat}} = [I, Z].$$

The actual observation factors there as

$$\boxed{\Lambda = A \Lambda_{\text{sat}}}, \quad (77)$$

where A is the original selected $r \times r$ column block.

On the full-rank locus,

$$\boxed{Q_\Lambda = A^{-*} Q_{\text{sat}} A^{-1}, \quad L_\Lambda = L_{\text{sat}} A^{-1}}. \quad (78)$$

Therefore its observed arithmetic action satisfies

$$\boxed{A_{B, \Lambda} = A A_{B, \text{sat}} A^{-1}}. \quad (79)$$

The metric and action transform together. A coefficient pole of A^{-1} is not by itself a blow-up of the action norm in its own quotient metric.

At a rank-drop point the literal target is $\text{im } A$. If L_A is the full minimum section of A in Q_{sat} , the literal observed action is

$$A_{B, \text{literal}} = A A_{B, \text{sat}} L_A,$$

and the exact descent defect is

$$\boxed{A A_{B, \text{sat}} - A_{B, \text{literal}} A = A A_{B, \text{sat}} P_{\ker A}}. \quad (80)$$

Thus the new kernel is retained in the arithmetic action.

5.9.3 9.3 An explicit finite pivot construction controls every parameter branch

The singular factor A can itself be analysed without assuming a one-parameter path.

Blow up the projective direction of its list of entries. On a chart with selected entry a_1 , every entry is a_1 times its retained ratio. The selected ratio is one. Fixed permutations and elementary row and column operations reduce that ratio matrix to

$$\text{diag}(1, B_1).$$

Repeat on the entries of B_1 . After at most r pivots, each resulting chart has

$$A = P \text{diag}(d_1, \dots, d_r) Q, \quad d_i = a_1 a_2 \cdots a_i, \quad (81)$$

with P, Q holomorphic and invertible.

Each step is the closure of the explicitly displayed projective graph of the residual entries. Its charts cover every approaching parameter branch. This construction does not assert that the divisor of every a_i is already normal crossing.

All metric factors are retained:

$$P^* Q_\Lambda P = D^{-*} H_{\text{reg}} D^{-1}, \quad D = \text{diag}(d_i), \quad H_{\text{reg}} = Q^{-*} Q_{\text{sat}} Q^{-1}. \quad (82)$$

In particular,

$$\begin{aligned} \log \det Q_\Lambda &= \log \det Q_{\text{sat}} - 2 \log |\det P \det Q| \\ &\quad - 2 \sum_{j=1}^r (r - j + 1) \log |a_j|. \end{aligned} \quad (83)$$

The actual module discrepancy is

$$\text{coker } A \simeq \bigoplus_{i=1}^r \mathcal{O}/(d_i). \quad (84)$$

Along an analytic curve, $d_i = z^{\nu_i}$ times a unit, and every inverse-exterior exponent is the sum of the corresponding largest ν_i . The missing derivative orders remain in (84).

5.9.4 9.4 The boundary connection and all its mixing terms

In the holomorphic frame of (82), the Chern connection is

$$\Gamma = D H_{\text{reg}}^{-1} \partial H_{\text{reg}} D^{-1} - D' D^{-1}. \quad (85)$$

The second term is the explicit logarithmic factor

$$- \sum_j \text{diag}(0, \dots, 0, 1, \dots, 1) \frac{\partial a_j}{a_j}.$$

The first term retains every off-diagonal mixing entry, multiplied by its actual ratio d_i/d_j .

On the complement of the divisor,

$$\mathcal{F}_\Gamma = D \mathcal{F}_{H_{\text{reg}}} D^{-1}. \quad (86)$$

The map D^{-1} is the exact isometry in (82), so the curvature norm in that metric is the norm of the regular curvature. It is bounded on compact regular-metric charts.

This Chern connection is not silently the real metric connection used for the heat derivative. In a given frame their difference is exactly

$$\boxed{\nabla^{\text{Ch}} - \left(d + \frac{1}{2} G^{-1} dG \right) = \frac{1}{2} G^{-1} (\partial G - \bar{\partial} G).} \quad (87)$$

Equations (81)–(87) provide a finite parameter model with explicit weight factors, complete extension data, and a controlled regular remainder. They do not construct a Hodge filtration or a Frobenius action that is absent from the arithmetic comparison.

5.10 10. What this gives the original programme

The new control is uniform in the aspects that can be made uniform from the actual finite positive-source data:

- Every fixed original subquotient has the same complex logarithmic neighbourhood (5) and all-order mixed derivative bounds (9)–(10).
- All covariance scales and noncommuting source ranges have the exact filtration model (24)–(32), including every metric and action exterior exponent.
- The complete observed heat has the no-gap memory representation (35)–(42), and its comparison with the actual observed squared action is bounded by the original kernel rank in (45).
- Every proper rational testing family below the exact conductor degree bound is faithfully observed, with the original metric return (65)–(66), including repeated and colliding poles.
- Source-rank and observation-rank degenerations retain their full missing-value modules and additional minima in (75), (80), and (84).

For the original simple family, the bulk observed positive-heat comparison is now completed by (1)–(2). The joint-source positive spectral comparison is below the kq scale by (50)–(51). The kernel's low proper-rational complexity is excluded by (3), uniformly over pole choices in its stated arithmetic domain.

The remaining scalar values are still

$$\mathcal{R} \log \det(I_K^* G_N I_K),$$

at its unresolved finer coefficient,

$$\mathcal{M}^p = \mathcal{R} \log \det \left[(Q_N^p)^{-1/2} H_N^p (Q_N^p)^{-1/2} \right],$$

and the actual native projected-current sign. Their complete maps are retained here; they are not replaced by the new saturated or weighted metrics.

I am not claiming a fresh executed numerical checker receipt or an arithmetic zero computation for these new proofs. The exact finite examples above are included to make the identities and limiting scales directly checkable.

The finite-family result is now considerably closer to the kind of control you asked for: a common analytic domain, all mixed derivatives, explicit boundary weights, complete source and observation extensions, and uniform heat-memory bounds. The uncontrolled native quantity is no longer hidden inside an unspecified change of metric or disappearing source direction; it remains as the explicit original determinant or phase evaluated through those maps.